

Seven year remediation study at the Carrie Bay Atlantic salmon (*Salmo salar*) farm in the Broughton Archipelago, British Columbia, Canada

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Abstract

Numerous studies describing the short-term recovery of benthic habitats impacted by organic wastes from commercial salmon farms have been published. A few of these studies have continued for up to 2 years. The Carrie Bay salmon farm in the Broughton Archipelago of British Columbia began operation in 1990 and produced four crops of Atlantic salmon. It was fallowed in 1998 after producing 2587 mt of Atlantic salmon in the preceding production cycle. Significant accumulations of organic waste were observed near the netpens in 1997 when the farm was close to maximum salmon biomass. Bottom sediments were blackened with iron sulfides and gas ebullition together with extensive *Beggiatoa* mats were observed to a distance of 145 m on the southern transect in 1997. Patchy *Beggiia* remained near the netpens in 1999. Twelve benthic surveys have been completed in Carrie Bay over the last 7 years. The database reported herein describes slow, but steady, remediation of this site. Chemical remediation, including reductions in total volatile solids (TVS), free sulfides, redox potential and zinc were nearly complete at distances >80 m in 2002 and rapid and significant additional chemical remediation was observed throughout the area in 2003. Sediment concentrations of zinc have declined throughout this study at all distances and were approaching background concentrations in 2002 and 2003. Macrofaunal communities at Carrie Bay's reference stations were dominated by mollusks with significantly fewer annelids and very few crustaceans. This community did not respond to organic enrichment in the classic way described by Pearson and Rosenberg [Oceanogr. Mar. Boil. Annu. Rev. 16 (1978) 229], which has accurately predicted the macrobenthic response at many intensive aquaculture facilities. None of the eight opportunistic taxa frequently found proliferating in enriched sediments near B.C. salmon farms were a large component of the Carrie Bay macrobenthos at any place or at any time. Rather, a single annelid (*Nephtys cornuta*) was found in moderate abundance and several bivalve mollusks that

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dominated local reference communities have slowly recruited into the remediating sediments. Biological remediation was nearing completion at distances ≥ 80 m from the netpens, but was not complete under and at shorter distances in 2002 following 4 years of fallow. Based on regression analysis, it is predicted that chemical remediation (sulfides ≤ 447 μM) sufficient to support half of the common taxa observed at the local reference station will be complete in 2004 following approximately 65 months in fallow. Unlike four other farms in the Broughton Archipelago, where sulfide concentrations exceeded 20,000 μM , and where chemical remediation has been complete in less than 6 months, sulfide concentrations at Carrie Bay have remained < 9410 μM throughout this study. These relatively lower sulfide concentrations; the unique pattern of biological remediation; and the relatively long chemical and biological remediation times observed at Carrie Bay point out the need for a better understanding of the diagenesis of labile carbon compounds in sediments underlying intensive aquaculture operations. A better understanding of the underlying causes of the biochemical processes observed in this study will significantly aid farm operators and regulators in selecting benthic habitats with optimal sustainability for modern salmon farming. In addition to describing the transport of waste, computer programs designed to predict the environmental response to intensive aquaculture must incorporate modules assessing the biogeochemical and biological response to these inputs before their output can predict the unusual responses observed in Carrie Bay.

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1. Introduction

Brooks et al. (2003) reviewed the literature describing benthic remediation under and adjacent to Atlantic salmon (*Salmo salar*) farms worldwide and described chemical remediation through several successive production cycles at four highly productive British Columbia salmon farms that had all been in operation for at least 10 years. In each case, the farms were fallowed for short periods (4 to 6 months) between harvest and the introduction of a new year-class. He defined the following terms.

Chemical remediation is the reduction of accumulated organic matter with a concomitant decrease in free sediment sulfide ($\text{S}^{=}$) concentrations and an increase in sediment redox potential under and adjacent to salmon farms to levels at which more than half the reference area taxa can recruit and survive.

Biological remediation is defined as the restructuring of the infaunal community to include those taxa whose individual abundance equaled or exceeded 1% of the total invertebrate abundance at local reference stations. Recruitment of rare species representing $< 1\%$ of the reference abundance is not considered necessary for complete biological remediation.

Based on an analysis of concentrations of free sediment $\text{S}^{=}$ (H_2S , HS^-), Brooks et al. (2003) predicted chemical remediation times of ≤ 6 months following harvest at the four Broughton Archipelago farms examined. These results were similar to the 185 day recovery, based on benthic respiration rates, reported by Woodward et al. (1992). Short chemical and biological benthic remediation times have also been reported by Ritz et al. (1989) in Tasmania; Brooks (1993) in Washington State and Arrow Pass in British

Columbia (Brooks et al., 2003). Anderson (1992) studied benthic effects at eight salmon farms in British Columbia and reported substantial recovery from moderate impacts in times as short as 6 months. Significant benthic recovery was observed to occur within 1 month at sites with comparatively high current speeds.

In contrast, Mahnken (1993) reported a longer remediation time adjacent to a salmon farm in Clam Bay, Washington that was located in a depositional environment. During fallow, he observed a 3-month period of rapid biological recovery with large increases in macrofaunal abundance and diversity followed by a second period lasting 25 months in which community recovery proceeded more slowly. Species that were rare at the reference location remained absent from the Clam Bay site at the end of his study. Morrissey et al. (2000) described the accumulation and catabolism of salmon farm waste at a New Zealand chinook salmon farm where a maximum current speed of 15 cm/s was recorded and where the minimum 2-h average speed of bottom currents was 1.1 cm/s. Sediments consisted of 33% to 70% mud at this shallow (26 m water depth) depositional site where organic enrichment was restricted to ≤ 50 m from the netpen's perimeter. Waste accumulated to depths of 27.5 to 43.1 cm under the farm's footprint and mean calculated remediation times, based on observed carbon catabolic rates varied between 3.3 and 7.5 years. Similarly, Karakassis et al. (1999) reported long chemical and biological remediation times of greater than 23 months at a shallow (19 m depth), depositional (71–72% silt and clay) Grecian site that had been annually producing a small quantity (140 mt) of sea bream and sea bass during the preceding 6 years. The physicochemical results of their study were compounded by significant blooms of autotrophs (benthic diatoms), which would not be expected to flourish in the deeper waters (>30 m) characterizing the enriched footprint at the salmon tenures studied in British Columbia. They reported an abundance of the carbon tolerant opportunist, *Capitella* cf. *capitata*, on the perimeter of the farm during the entire 23-month study. The abundance of this opportunist diminished at the 10 m station following 8 months of fallow.

Morrissey et al. (2000) reported mean (± 1.0 standard error) sediment zinc concentrations of 665 ± 19.5 $\mu\text{g Zn/g}$ dry sediment under a salmon farm's footprint. The authors did not report temporal changes in zinc concentrations during chemical remediation. They noted that the bioavailability of zinc is decreased in anaerobic conditions and assumed that it would be released from organic carbon and sulfide bound phases resulting in increased toxicity that could inhibit biological recovery. In contrast, Brooks and Mahnken (2003a,b) reported that initially high (200 $\mu\text{g Zn/g}$ dry sediment) concentrations at the Arrow Pass salmon farm in British Columbia declined to background concentrations (25 $\mu\text{g Zn/g}$) during 11 months of chemical remediation and had no observable affect on biological remediation.

These reports indicated that benthic recovery times vary significantly from a few months at some sites to a several years at heavily impacted sites. It should be noted that all of these reports described remediation under intensive aquaculture facilities located in depositional environments. Brooks and Mahnken (2003a) and Nash (2001) reported significantly increased diversity and abundance of macrofauna under and downcurrent from Pacific Northwest salmon farms located over erosional environments characterized by fast currents (up to 125 cm/s) and coarse sediments. These results suggest that remediation times are proportional to at least the depth of accumulated

waste, sediment and water physicochemical characteristics and water depths and current speeds.

2. Background

Stolt Sea Farm obtained a salmon aquaculture lease in Carrie Bay during 1990 (Fig. 1). Four crops of Atlantic salmon were raised at the farm prior to an extended fallow period, which began in August 1998. During the last of four production cycles, a total of 3881 mt of feed were used to produce 2587 mt of Atlantic salmon, which were harvested in July 1998. The producer initiated a benthic survey of the dominantly downcurrent southern transect in early 1997 when the farm was near peak biomass. The survey revealed a significant accumulation of waste on the netpen's perimeter. Sediments were black with iron sulfide, smelled strongly of hydrogen sulfide and were covered with patchy mats of the sulfide oxidizing bacterium *Beggiatoa*. These effects extended, in varying degrees to a distance of between 145 and 185 m south of the netpen's perimeter.

3. Methods

A regression approach was used throughout these studies. A dominantly down-current vector (196 °T) was identified based on video surveys; a 15 m depth current meter study carried out over one full lunar cycle; and a drifter study (Sea Consult Marine Research, 1997). Single sediment samples were collected along south bearing transects at 15 to 30 m intervals to a distance of 225 m and three replicate reference station samples were collected during most sample periods at hydrologically isolated locations >500 m from the farm. Three additional transects (023, 113 and 293 °T) were examined in July 1999 to determine the extent of the farm's effects and to confirm that the southern transect was most affected. Assessment methods at this site have included periodic video records coupled with physicochemical and biological surveys. Surveys were conducted at least annually in the summer or fall. Five surveys were completed in 2000 (February, April, June, August and October). Physicochemical and macrofaunal surveys were undertaken in August to October of 2000, 2001 and 2002 to minimize seasonal variation. The last physicochemical survey was completed in April 2003. These surveys are summarized in Fig. 2 and Table 2.

3.1. Station positioning

Station positioning prior to netpen removal was determined via a 12.7 mm polypropylene line attached to the midpoint of the cages. The other end of the line was secured to a cleat adjacent to the sampling station. The boat was then maneuvered to place the grab over the appropriate magnetic bearing with the transect line held tight. Differential GPS was used, beginning in 1999, to position the research vessel following

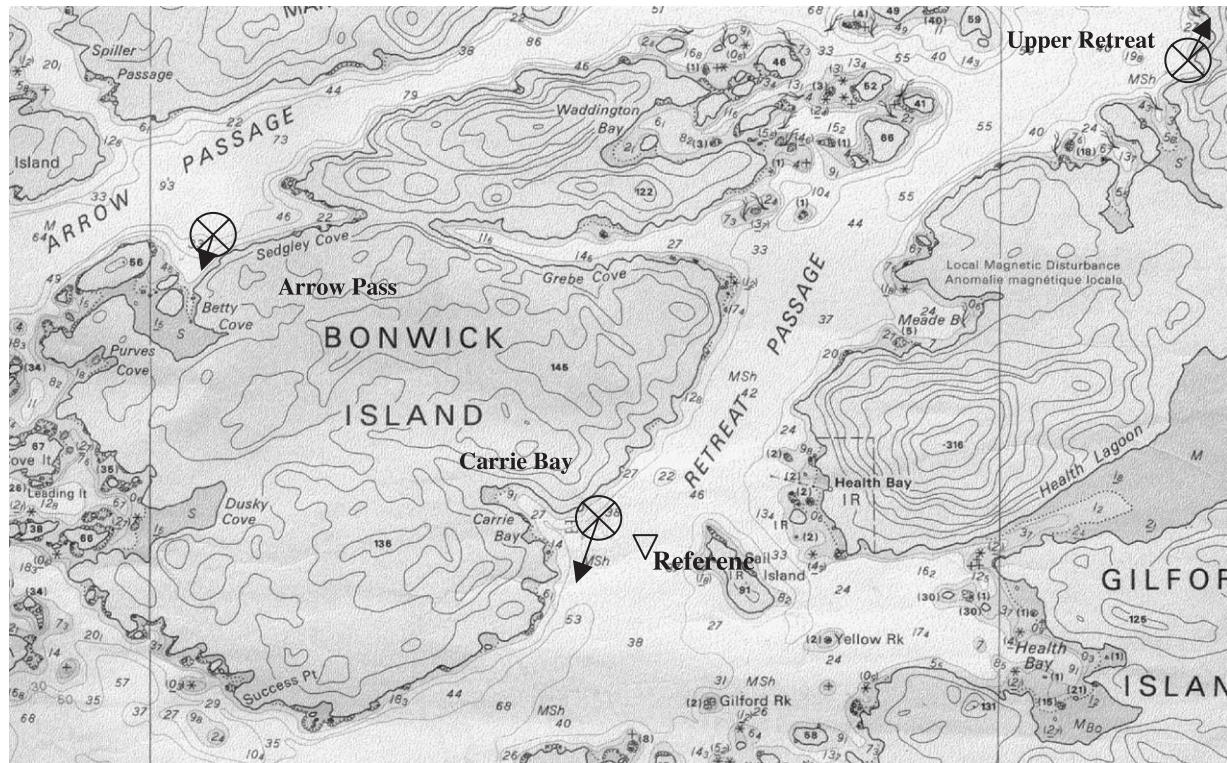


Fig. 1. Site map describing the locations of the Carrie Bay, Upper Retreat and Arrow Pass salmon farms in the Broughton Archipelago of British Columbia. The 15 m deep direction and speed of the net current vector at each farm is indicated.

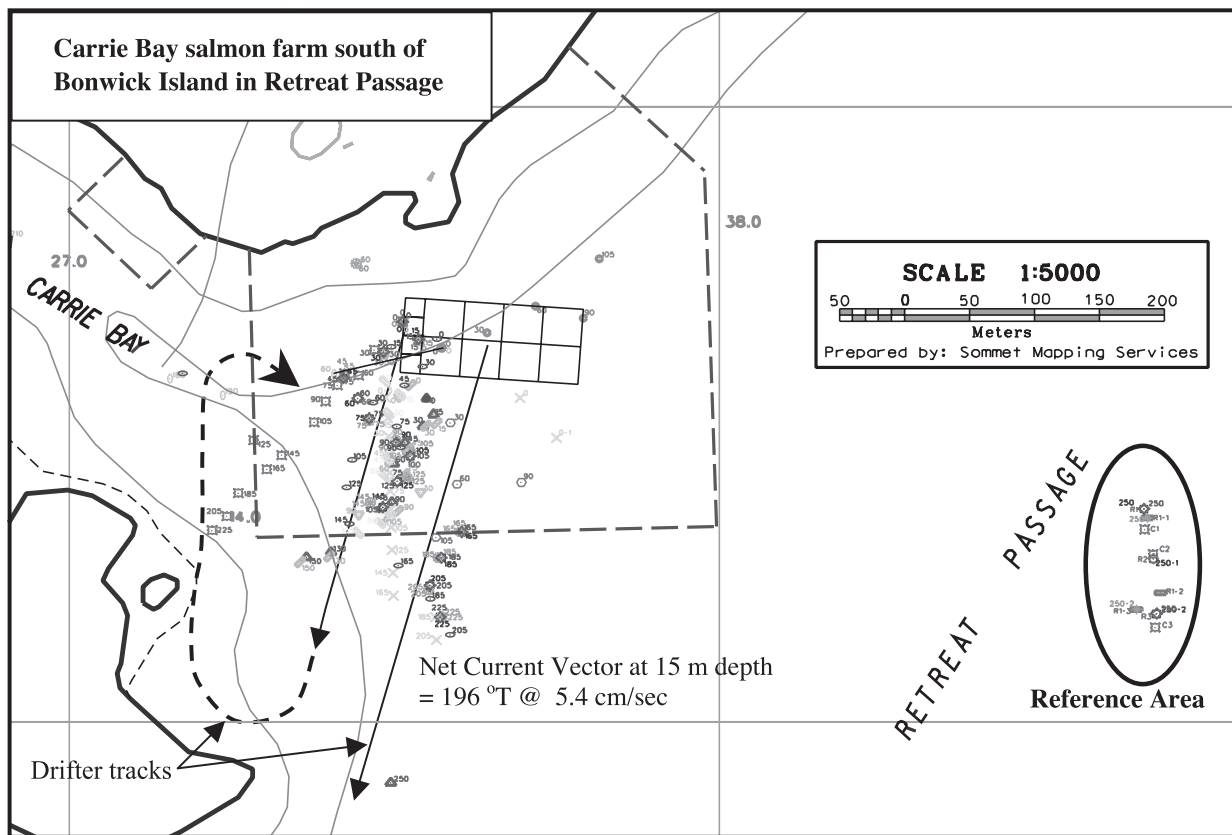


Fig. 2. Carrie Bay salmon farm followed in August 1998. Current vectors observed in drifter and current meter studies are indicated together with the location of each of the 161 sediment samples collected during a remediation study lasting from 1997 through 2003. [Table 2](#) provides a key to transects and sample stations evaluated during each sampling event.

removal of the farm structure. All GPS locations were entered into GIS software and the distance from the nearest netpen perimeter determined analytically from the known position of the farm cages during the 1996–1998 production period. The location of each sample is provided in Fig. 1.

3.2. Sample collection

Sediment samples for physicochemical analyses were collected using a Petite Ponar grab with a footprint of 0.0225 m². The upper 2 cm of the entire Petite Ponar grab sample was used for physicochemical analyses as these smaller samples were not used for macrofaunal evaluations. Seventeen samples for physicochemical and macrofaunal analyses were collected with a stainless steel 0.1 m² van Veen grab on each sample day in 2000, 2001 and 2002. Acceptable samples met the requirements of the Puget Sound Estuary Program (PSEP, 1996), viz:

- The sampler was deployed at a maximum speed of 30 cm/s.
- A minimum sediment penetration depth of 4 cm was required.
- The sampler contained overlying water, with low turbidity, indicating minimal leakage.
- The retained sediment surface was relatively flat indicating minimal disturbance.

Overlying water was siphoned from the sampler through a 500 µM sieve to retain suspended organisms. Surficial sediments (top 2 cm) for physicochemical analyses were removed from not more than 5% of the surface area of the 0.1 m² van Veen grab (PSEP, 1996). Infaunal samples were placed in 5-gal buckets for processing at a shore station on 1.0 mm sieves on the day of collection.

3.3. Sediment core samples

In April 2003, sediment cores were collected from van Veen grab samples using 3.8 cm inside diameter schedule 40 PVC pipe. The ends of each 15 cm tube were sealed with PVC caps, marked to designate the sediment's upper surface and frozen at −20 °C. The sealed core tubes were placed under running cold water for 2 min and the still frozen cores extracted using a plexiglass plunger. The length of the core was measured and the RPD was determined prior to sectioning them into either 1.5 or 2.0 cm deep sections beginning at the sediment's surface. Total S[−] and TVS were measured in each section to assess these parameters as a function of depth beneath the sediment–water interface.

3.4. Total volatile solids (TVS)

TVS analysis was accomplished on 50-g surficial sediment samples collected from the top 2 cm of the grab (PSEP, 1996). Samples were dried at 103±2 °C in aluminum boats that had been tared following combustion at 550 °C for 30 min. Drying was continued until no further weight reduction was observed (generally overnight). The

samples were then combusted at 550 °C for 2 h or until no further weight loss was recorded. TVS were calculated as the difference between the dried and combusted weights. Triplicate analyses were completed on 5% of the samples or on a minimum of one per batch (PSEP, 1996).

3.5. Sediment grain size (SGS) analysis

SGS was accomplished during the baseline survey and annually thereafter using the method of Plumb (1981). The sediments were wet sieved on a 0.064 mm sieve. The fraction retained on the sieve was dried in an oven at 92 °C and sieved on mesh openings of 2.0, 0.89, 0.25, and 0.064 mm. Particles passing the 0.064 mm sieve during wet sieving were analyzed by sinking rates in a column of water (pipette analysis). Triplicate analyses were completed on 5% of the samples with a minimum of one per batch.

3.6. Reduction–oxidation potential discontinuity (RPD)

Prior to 1999, sediment redox potential was assessed by measuring the depth of the oxidation–reduction potential discontinuity (RPD). A clear 2-cm diameter corer was inserted into the undisturbed sediment in samples collected for physicochemical analysis. The depth at which the sediment color changed from the background (gray-green) color to black (indicative of iron sulfides associated with anaerobic conditions) was measured in cm.

3.7. Benthic infaunal analysis

The entire contents of the 0.1 m² infaunal grab samples (including the overlying water) were washed from the grab into a 5-gal bucket using 50 µm filtered seawater. Samples were then sieved on 1.0 mm stainless steel screens. The retained material was placed in 1.0 or 2.0 l HDPE bottles and fixed using 15% buffered formalin in seawater on the day of collection. Each sample jar had matching inside, outside and cap labels. Fixed samples were washed with filtered fresh water and preserved in 70% isopropyl alcohol after 4 days. Infaunal organisms were sorted from the background matrix under 10× magnification. A different technician repicked 20% of each sample. Quality assurance guidelines required a picking efficiency of ≥95%. Any sample failing this QA benchmark was completely repicked. Infauna was identified to the lowest level practicable—generally to species. All taxa were compared with verified specimens in a reference collection (Aquatic Environmental Sciences), or will be verified by an outside expert.

3.8. Macrofaunal wet tissue weights

All annelids were removed from their tubes, which were not included in the weights. The shells of living mollusks were left intact. Wet tissue weights were determined for macrofauna preserved in 70% isopropyl alcohol. The samples were

sieved on tared 100 μm Nytex™ screens. The bottom of the sieve was blotted with absorbent tissue and allowed to dry for 60 s prior to weighing on an electronic balance to the nearest 0.1 mg.

3.9. Sediment zinc

Sediment zinc concentrations were determined by Analytical Resources in Seattle, Washington using ICP (EPA Method 6010) following strong acid digestion. Quality assurance included the concurrent determination of spiked samples (75–125% of spike recovered), methods blanks (analyte not detected) and duplicates (one per run with $\leq 20\%$ relative percent deviation) as specified in PSEP (1996).

3.10. Redox potential (Eh)

Beginning in 1999, redox potential was measured in the field using an Orion™ advanced portable ISE/pH/mV/ORP/temperature meter model 290A with a Model 9678BN Epoxy Sure-Flow Combination Redox/ORP probe. The stated accuracy of the meter in the ORP mode is ± 0.2 mV or $\pm 0.05\%$ of the reading, whichever is greater. Detailed protocols for preparation of standards and conducting the analyses are provided in Wildish et al. (1999). Surficial sediments (top 2.0 cm) were gently homogenized and placed in new 125 ml urine specimen jars with no headspace. These were immediately placed in a cooler and held on shaved ice until the samples were analyzed simultaneously for pH, Eh and sulfides. The samples were allowed to warm to ca. 10 °C (the approximate temperature of the sediments) prior to removal of 5 ml of sediment for the sulfide analyses. The redox probe, which was positioned on a ring-stand, was inserted approximately 2.0 cm into the sample container, which was gently agitated to insure that no air remained around the probe and that it had intimate contact with the sediments. The probe was then allowed to stabilize prior to taking the reading. All onboard analyses were completed within 30 min of collecting the samples. Quality assurance required checking of the meter against two standards at the beginning and end of each batch of samples and triplicate analyses completed on 5% of the samples with a minimum of one per batch of 20 samples.

3.11. Free sediment sulfides (S^-)

Prior to 1999, S^- were evaluated organoleptically on a scale of zero (no detected hydrogen sulfide) to five (very strong and lingering smell of H_2S). Beginning in 1999, free sediment S^- were evaluated using an Orion™ advanced portable ISE/pH/mV/ORP/temperature meter model 290A with a Model 9616 BNC Ionplus Silver/Sulfide electrode. The meter has a concentration range of 0.000 to 19,900 μM and a relative accuracy of $\pm 0.5\%$ of the reading. The probe and meter were three point calibrated using 100, 1000 and 10,000 μM standards every 3 h. Detailed protocols for the preparation of standards are provided in Wildish et al. (1999). Quality assurance required frequent checking of the meter against standards and triplicate analyses completed on 5% of the samples with a minimum of one per batch. Sulfide analyses were completed using 5 cm^3 subsamples of

the homogenized surficial sediment samples described above. Sediment samples were held on shaved ice until analyzed for sulfides within 30 min of collection.

3.12. Underwater video

A cabled Outland Technology™, UWC-160 camera, equipped with a UWL-22 (200 W) 115 V light was used to record video on a Sony combination TV-VCR. The camera and light were attached to the control unit with 100 m umbilical cord. The unit was housed in an aluminum frame which included a large stabilizing fin and a “T” probe marked in 5 cm increments to establish horizontal dimensions. The field of view at the length of the probe was 0.5 m. The unit was towed across the survey area at an average speed of 0.25 m/s. Camera position was recorded as a function of tape time using differential GPS with no correction made for hydrowire angle.

3.13. Statistical analyses

The experimental design and analyses relied primarily on correlation, principle components and linear and nonlinear regression analyses. Raw data was entered into a Microsoft Excel™ spreadsheet and imported into Statistica™ software for analysis. Proportional data (TVS and SGS) were arcsine(square-root) transformed prior to inferential analyses (Zar, 1984). Biological count data were transformed ($\ln(N+1)$) prior to determination of Pearson correlation coefficients. Biological responses were summarized in three-dimensional contour graphs using Statistica's distance weighted least squares subroutine. The location in time and space of each sample is identified in each of these graphs. Predictions outside the range of these sample locations must be interpreted with caution because they are based on trends and not on actual measured values. Statistical significance throughout this report is associated with the probability of making a Type I error of 0.05.

Shannon's Index (Shannon and Weaver, 1949). This index provides the average uncertainty per species in an infinite community of taxa. The form of the index used in this analysis is:

$$\text{Shannon's Index} = H' = - \sum_{\text{over species}} (n_i/n) \ln(n_i/n)$$

The value of the index is zero when a single species is present. It is maximized when there are a large number of equally represented species and it is reduced in communities dominated by a few highly abundant species. The value of Shannon's Index in a sample containing 20 taxa equally represented in a total abundance of 600 animals would be 3.0.

Pielou's Index (Pielou, 1977) is a commonly used measure of community evenness. It expresses the observed value of Shannon's Index relative to the maximum possible value ($\ln(S)$) where (S) is the number of taxa present. Pielou's index varies between 0 and 1.0 and generally covaries with Shannon's Index.

$$\text{Pielou's Index} = J' = H' / \ln(S).$$

4. Results

Fig. 1 describes the location of the Carrie Bay salmon farm in the Broughton Archipelago of British Columbia. The location of the Arrow Pass and Upper Retreat tenures, used for comparison in this report, are also displayed along with arrows defining the net transport vectors at each farm.

4.1. Site characterization

Selected hydrodynamic and benthic physicochemical properties observed at Arrow Pass, Upper Retreat and Carrie Bay are summarized in Table 1. Water depths are similar at all three farms. Carrie Bay sediments had a significantly higher proportion of fines (silt and clay=71.5% to 92.4%) suggesting slow bottom current speeds and a more depositional benthos that was found at Upper Retreat or Arrow Pass. Total volatile solids were higher at Carrie Bay's reference stations (9.2% to 11.2%) than at either Upper Retreat (7.7% to 9.05%) or Arrow Pass (4.1% to 5.5%).

Carrie Bay's 15 m deep mean current speed (5.9 cm/s) and net current vector (5.4 cm/s at 196 °T) suggests that waste was preferentially carried to the south. However, the 30 m deep drifters released near the inner (west) end of the farm by Sea Consult (1997) were deflected into a clockwise gyre as they approached the southern shore of the bay. This apparent gyre concentrated flotsam in the bay, which was frequently confirmed by farm workers. Drifters released near the eastern end of the netpen array proceeded south down Retreat Passage with no deflection.

Fig. 2 describes the location of the farm, the differential GPS position of each of the 161 sediment samples collected during these surveys and the primary current vectors defined by the drifter and current meter studies and Table 2 summarizes the surveys by date. The farm's footprint was assessed on July 14, 1999 following 1 year in fallow, by determining TVS, $S^{=}$ and redox potential in sediments at ~30 m intervals from the center of the pre-existing cage system on four magnetic bearings. Consistent with the hydrodynamic data, the concentrations of $S^{=}$ and TVS, summarized in Table 3, describe a waste deposition pattern after 1 year in fallow that was skewed to the south along the 196 °T net transport vector, but which also extended westward from the western netpen array's perimeter into the bay a distance of 60 m based on the $S^{=}$ data or to over 105 m based on the TVS record. The following assessment of chemical and biological

Table 1

Summary of physicochemical and hydrodynamic conditions at the Arrow Pass, Upper Retreat and Carrie Bay salmon farms

Farm	Depth (m)	% Fines	% Sand	Reference, % TVS	Current speeds (cm/s)		
					Mean	Max.	Net current vector
Arrow Pass	47.7–56.0	13.1–35.5	52.1–77.9	4.09–5.52	3.4	25.0	2.0 (193 °T)
Upper Retreat	36.5–37.7	38.6–63.0	36.4–60.8	7.73–9.05	4.8	25.0	2.2 (036 °T)
Carrie Treatment	34.1–39.0	71.5–92.4	7.6–26.4	9.19–11.20	5.9	25.0	5.4 (196 °T)
Carrie Reference	41.2–45.2	68.3–75.0	24.6–29.4	9.19–11.20			

Table 2
Sampling dates with a key to the sample locations provided in Fig. 2

Sample date	Symbol	Intended true bearing	Number & type of samples	Sample location or Sample interval & transect end	Analyses completed
11/14/1997	No GPS	196	6 treatment and 2 control	0, 30 & 100 m	TVS, Zn, SGS, Video
11/19/1998	No GPS	245	5 treatment	0, 30, 50 & 70 m	TVS, Zn, SGS
05/17/1999		196	8 treatment	15 m to 105 m	TVS, S ⁼ , Zn, SGS
07/14/1999		196	10 treatment and 3 control	15 m to 105 m plus 130 & 150 m	TVS, S ⁼ , Zn, Video
07/14/1999		023	5 treatment	30 m to 120 m	TVS, S ⁼ , Video
07/14/1999		113	5 treatment	30 m to 120 m	TVS, S ⁼ , Video
07/14/1999		293	5 treatment	60 m to 210 m	TVS, S ⁼ , Video
02/24/2000		196	11 treatment	15 m to 105 m plus 130, 150 & 250 m	TVS, S ⁼
04/22/2000	No GPS	205	11 treatment	15 m to 105 m plus 130, 150 & 250 m	TVS, S ⁼
06/13/2000		205 plus perimeter stations	3 treatment	0, 30 & 90 m	TVS, S ⁼ , SGS
08/23/2000		196	16 treatment and 3 control	15 m to 105 m & 20 m from 105 to 205 m	TVS, S ⁼
10/16/2000		196	14 treatment and 3 control	15 m to 105 m & 20 m from 105 to 225 m	TVS, S ⁼ , SGS, Macrobenthos
07/31/2001		196	14 treatment and 3 control	15 m to 105 m & 20 m from 105 to 225 m	TVS, S ⁼ , Zn, SGS, Macrobenthos
10/25/2002		196	14 treatment and 3 control	15 m to 105 m & 20 m from 105 to 225 m	TVS, S ⁼ , SGS, Macrobenthos
05/08/2003	No GPS	203	14 treatment and 3 control	15 m to 105 m & 20 m from 105 to 225 m	TVS, S ⁼ , Zn, SGS TVS cores

The analyses reported herein are total volatile solids (TVS), free sulfides with an ion specific probe (S⁼), sedimented zinc (Zn), sediment grain size (SGS), Macrobenthos and Video.

Table 3

Summary of (a) sediment free sulfides (μM) and (b) TVS (percent) observed in sediments on four transects from the Carrie Bay salmon farm on July 14, 1999

Transect ($^{\circ}\text{T}$)	0	30	60	90	105	140	160	Reference ($N=3$)
<i>(a) Free sediment sulfides (μM)</i>								
023	3090–4070	162	213	–	–	–	–	114–184
113	9410	8940	405	14	68	–	–	114–184
196	5510	8220	821	522	318	549	93	114–184
293	3420	–	344	–	213	–	222	114–184
<i>(b) Percent sediment TVS</i>								
023	12.7–14.1	9.3	10.0	–	–	–	–	9.2–10.6
113	17.8	21.4	11.7	10.6	10.8	–	–	9.2–10.6
196	19.8	20.7	13.5	12.6	12.2	11.7	4.2	9.2–10.6
293	17.8	–	14.6	–	12.9	–	11.2	9.2–10.6

remediation in sediments will focus on samples collected along southern transects at Carrie Bay between 1997 and 2003.

4.2. Reference station physicochemical and biological characterization

Summary statistics for physicochemical and macrobenthic variables are provided in Table 4 for the reference stations used when macrobenthic samples were collected. Reference sediments in this area had a mean redox potential that was negative (-5.38 mV). That likely reflects the fine-grained nature of their sediment structure ($73.0 \pm 3.1\%$ silt and clay) and their high organic content ($10.04 \pm 0.86\%$ TVS). Brooks and Mahnken (2003a) reported that Total Organic Carbon (TOC) equaled $0.629(\text{TVS} - 0.009)$ in same sample analyses at the Arrow Pass farm. The constant and coefficient on TVS were both

Table 4

Summary statistics for primary (P) reference sediments collected in 2000, 2001 and 2002 near the Carrie Bay salmon farm

	Mean	–95% CI	+95% CI	Minimum	Maximum
Depth (m)	43.30	41.91	44.69	41.2	45.20
TVS (%)	10.04	9.18	10.90	9.19	11.20
Sulfide (μM)	34.17	16.97	51.37	9.17	53.70
Redox potential (mV)	–5.38	–75.24	64.47	–79	101.57
Zinc ($\mu\text{g Zn/g}$)	63.33	60.46	66.20	62.00	64
Gravel (%)	1.46	0	3.83	0.19	6.05
Sand (%)	25.68	23.74	27.63	24.57	29.35
Fines (%)	72.97	69.87	76.07	68.29	74.97
Total abundance/ m^2	2848	1896	3801	1690	3980
Number taxa/ 0.1 m^2	35.33	29.75	40.91	28.00	41.00
Shannon's Index	2.38	2.08	2.67	1.89	2.63
Pielou's Index	0.67	0.61	0.72	0.57	0.72
Number annelids/ m^2	702	331	1073	290	1280
Number mollusks/ m^2	2098	1453	2743	1130	2.780
Number crustaceans/ m^2	42	21	62	10	60

Table 5

Summary statistics for all Carrie Bay treatment station samples collected between 1997 and 2003

	Mean	−95% CI	+95% CI	Minimum	Maximum
Depth (m)	36.53	35.73	37.33	22.62	49.50
TVS (%)	13.07	12.38	13.76	4.20	24.28
Sulfide (μM)	1349	1003.4	1693.69	13.50	9410
Redox potential (mV)	−123.57	−133.03	−114.11	−249.45	−1040
Zinc ($\mu\text{g Zn/g}$)	144.66	103.51	185.81	18.00	419
Gravel (%)	1.86	−0.10	3.82	0.00	32.29
Sand (%)	22.78	19.53	26.04	7.55	58.85
Fines (%)	75.04	71.29	78.78	33.45	92.45
Total abundance/ m^2	1839	1328	2351	160	6320
Number taxa/ 0.1 m^2	21.90	17.74	26.07	5	71
Shannon's Index	2.09	1.95	2.24	1.13	3.37
Pielou's Index	0.72	0.70	0.75	0.52	0.90
Number annelids/ m^2	559	308	809	60	4840
Number mollusks/ m^2	1206	827	1584	40	5670
Number crustaceans/ m^2	60	31	88	0	440

significant and the coefficient of determination for the regression equaled 0.80. In this instance, reference sediments near Carrie Bay contained ca. 6.31% TOC. Microbial catabolism of this natural organic matter could cause the low redox. However, concen-

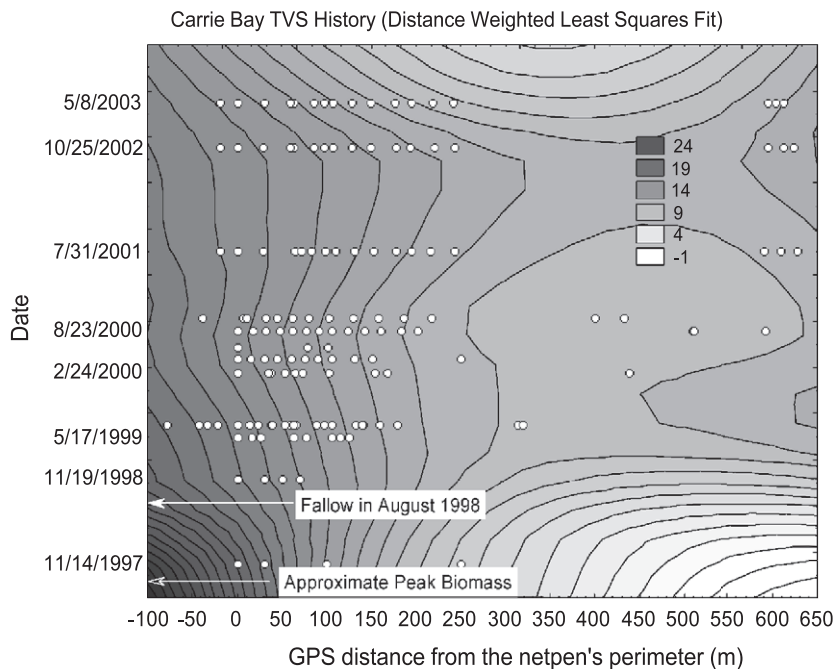


Fig. 3. History of sedimented TVS from near peak biomass in 1997 through May 2003 as a function of distance from the netpen perimeter and survey date. Percent TVS is plotted on the vertical (z) axis. The farm was fallowed in August 1998.

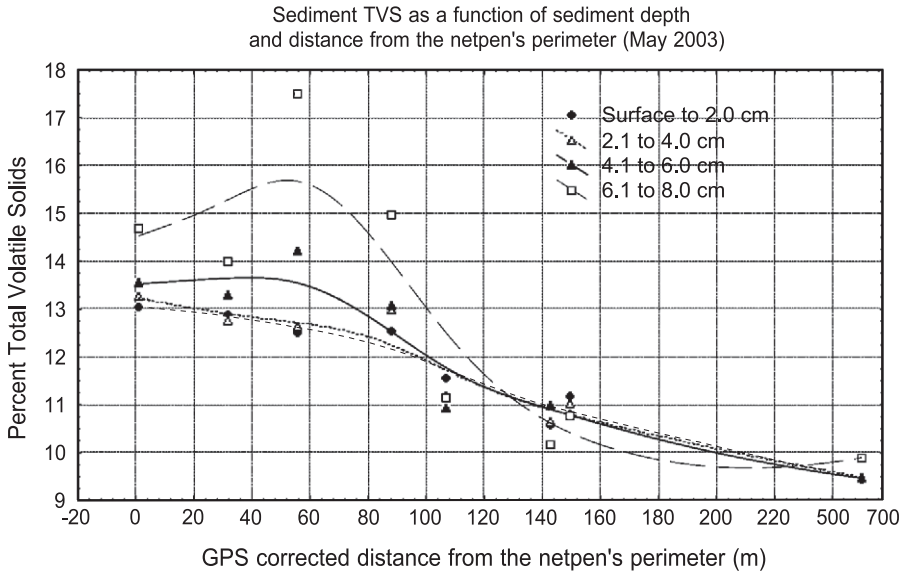


Fig. 4. Summary of TVS observed in sectioned core samples collected along the 203 °T transect from the netpen perimeter in 2003. Note the discontinuity in the ordinate's scale.

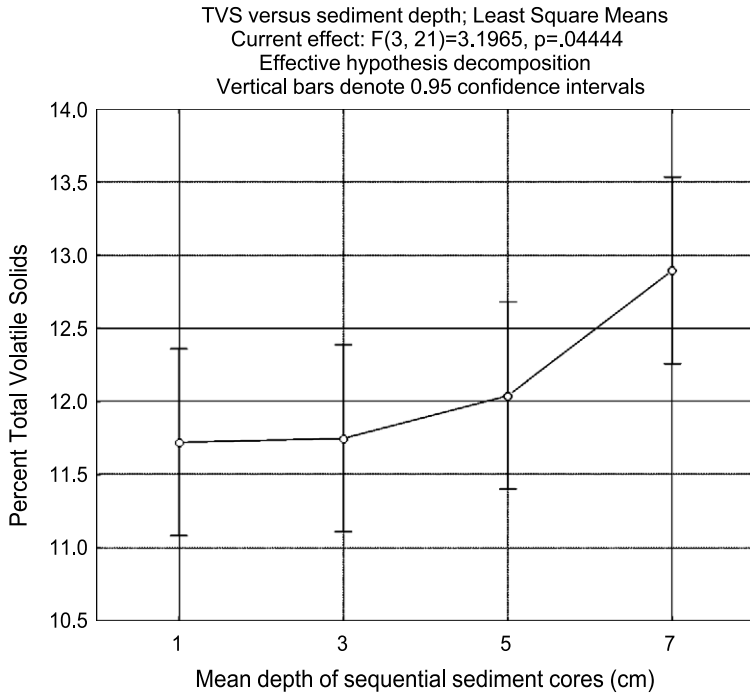


Fig. 5. TVS as a function of depth at farm stations located ≤ 150 m from Carrie Bay's netpen perimeter.

trations of $S^{=}$ in reference sediments were low $34.17 \pm 17.2 \mu\text{M}$. Brooks and Mahnken (2003a) found that $S^{=}$ were the most reliable indicator of biological response in sediments collected at eight British Columbia salmon farms and this endpoint will be emphasized in the analysis that follows.

Table 5 lists dominant taxa, representing $>1.0\%$ of the total abundance, observed in Carrie Bay's reference sediments. Mollusks accounted for 73.7% of the reference macrofaunal community and annelids 18.3%. Four crustacean taxa retrieved from these reference samples represented only 1.2% of the macrobenthos.

4.3. Sediment physicochemical history at Carrie Bay

4.3.1. Total volatile solids

Fig. 3 describes the TVS history along southern transects at Carrie Bay. The distance weighted least squares fit to the data suggests that accumulated TVS, which exceeded 20% under the footprint of the farm during production, was degraded and dispersed during the period of this study. Initially, biodeposits from the farm increased TVS to greater than the primary reference TVS of ca. 10% for a distance >100 m downcurrent. Patchy *Beggiatoa* was observed to 200 m on the southern transect in 1997 when the farm was near peak biomass. The 10% reference TVS contour increased to >200 m in 2002 as the organic burden under the farm (0 and minus values on the ordinate in Fig. 3) decreased from 24% to 13.5%. In 2003, background TVS concentrations had retreated to <205 m from the

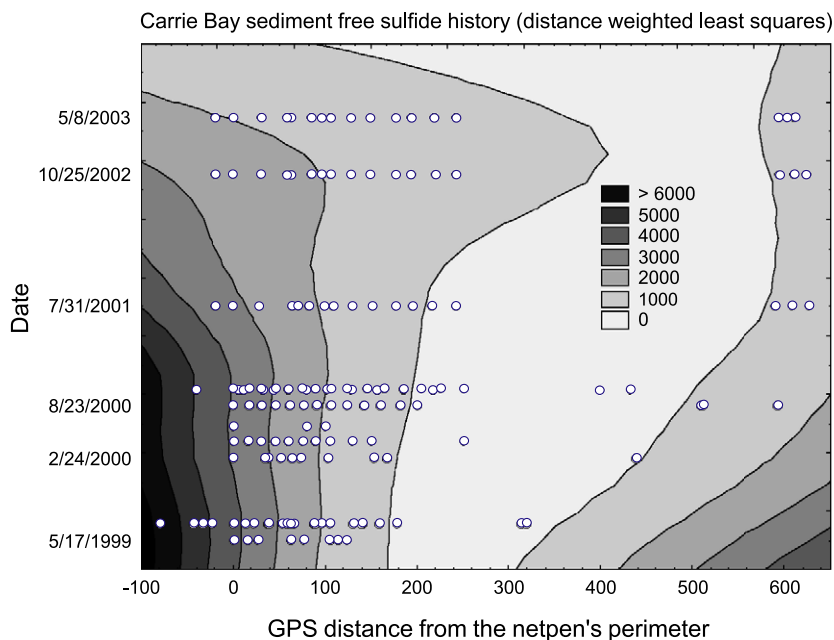


Fig. 6. Distance weighted least squares contour plot describing concentrations of free sediment sulfides (μM) near the Carrie Bay salmon farm between 1999 and 2003.

netpen's perimeter and observed TVS concentrations throughout the area were <12.8% suggesting that after 4 years in fallow, chemical remediation was proceeding slowly, but was not yet complete in the immediate vicinity of the farm.

4.3.2. Vertical sediment TVS profile

The RPD was clearly visible in Carrie Bay sediments. It increased in depth on the netpen's perimeter from 1.2 cm in 2000 to 3.1 cm in 2003. The RPD was deeper than 6.0 cm in all core samples collected at distances ≥ 30 m from the netpen's perimeter in May 2003. Fig. 4 summarizes the percent TVS observed as a function of distance and sediment depth in May 2003 when TVS concentrations in the top 4 cm of the sediment column were lower than observed in deeper sediments at distances less than 60 to 90 m from the netpen's perimeter. Significant differences, as a function of sediment depth, were found at all stations <88 m distance (MANOVA, $F=8.64$; $p=0.005$), but not as a function of distance from the netpen's perimeter ($F=1.08$; $p=0.41$). Post hoc testing (Duncan's Test) revealed that only sediments at depths >6.0 cm were significantly different from those lying above this horizon. Significant differences were not observed between the 0–2.0, 2.01–4.0 or 4.01–6.0 deep layers. ($p=0.20$ to 0.76). These differences are summarized in Fig. 5.

In contrast, beyond 88 m distance, significant differences were not found as a function of depth (MANOVA, $F=0.32$; $p=0.81$), but significant differences between all farm

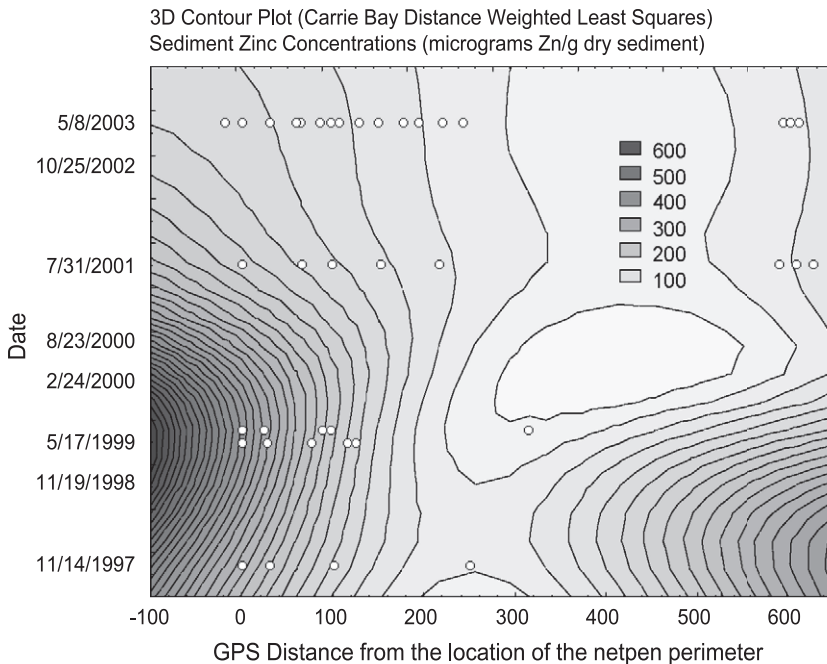


Fig. 7. Distance weighted least squares contour plot describing sediment zinc concentrations ($\mu\text{g Zn/g}$ dry sediment) as a function of distance from the locations of the Carrie Bay netpen perimeter between 1997 and 2003. The farm was fallowed in July 1998.

stations and reference conditions remained (ANOVA, $F=26.08$; $p<0.000$). Differences between farm stations were not significant beyond 88 m from the former netpen location.

4.3.3. Free sediment sulfides (S^-)

Fig. 6 summarizes the history of sediment S^- at Carrie Bay during this study. Sediment S^- was not measured until 1999 when concentrations as high as $9410 \mu\text{M}$ were observed on the netpen's perimeter. Concentrations remained high until October 2002 when the highest observed concentration was $2817 \mu\text{M}$. Consistent with the TVS record, S^- concentrations declined rapidly between 2002 and 2003 when the highest observed concentration was $1270 \mu\text{M}$ at 75 m. Beyond that distance, S^- remained slightly elevated in the range of 225 to $549 \mu\text{M}$ S^- .

4.3.4. Sedimented zinc

Concentrations as high as $351 \mu\text{g Zn/g}$ dry sediment on the perimeter of the Carrie Bay salmon farm and $269 \mu\text{g Zn/g}$ at 30 m were recorded in 1997. These values were largely elevated above the $63.3 \pm 2.9 \mu\text{g Zn/g}$ observed in reference sediments. Higher concentrations of $375.5 \pm 84.7 \mu\text{g Zn/g}$ ($N=4$) were observed in May and July 1999 at distances <30 m from the netpen's perimeter. Increased zinc concentrations ($\leq 158.0 \mu\text{g/g}$) were observed to the maximum distance assessed (105 m) in both years. Sediment zinc declined

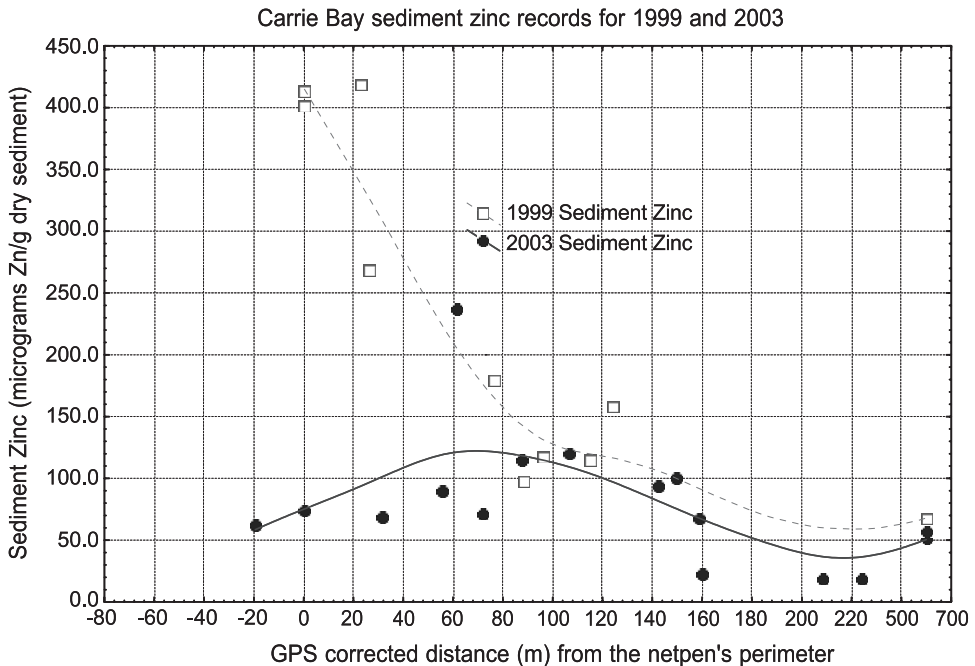


Fig. 8. Sediment zinc record for Carrie Bay in 1999 and 2003 following 5 years in fallow. Note the discontinuity in the ordinate. Reference samples were collected at 600 m from the netpen's perimeter on a bearing orthogonal to the prevailing currents.

to a maximum of 131 $\mu\text{g/g}$ at all stations in 2001 and background concentrations were observed at distances as close as 45 m. The entire zinc record for Carrie Bay sediments is summarized in Fig. 7 and the records for 1999 and 2003, which are compared in Fig. 8, emphasize the decline observed during this 4-year period. Excepting one high value in 2003, the record indicates that sediment zinc concentrations had declined to less than the zinc biological effects benchmarks reviewed by Brooks and Mahnken (2003b) along the most impacted transect at Carrie Bay.

4.3.5. Lateral transport of TVS

Cromey et al. (2002) introduced the DEPOMOD model for predicting the transport and fate of waste originating on salmon farms. Their model predicted that farm waste could be resuspended and laterally transported by benthic boundary layer current speeds ≥ 9.5 cm/s. Bottom current speeds were not measured at Carrie Bay and the high proportion silts and clays suggest that this is a depositional area with low bottom current speeds. However, current speeds frequently vary harmonically in tidally driven systems such as found at

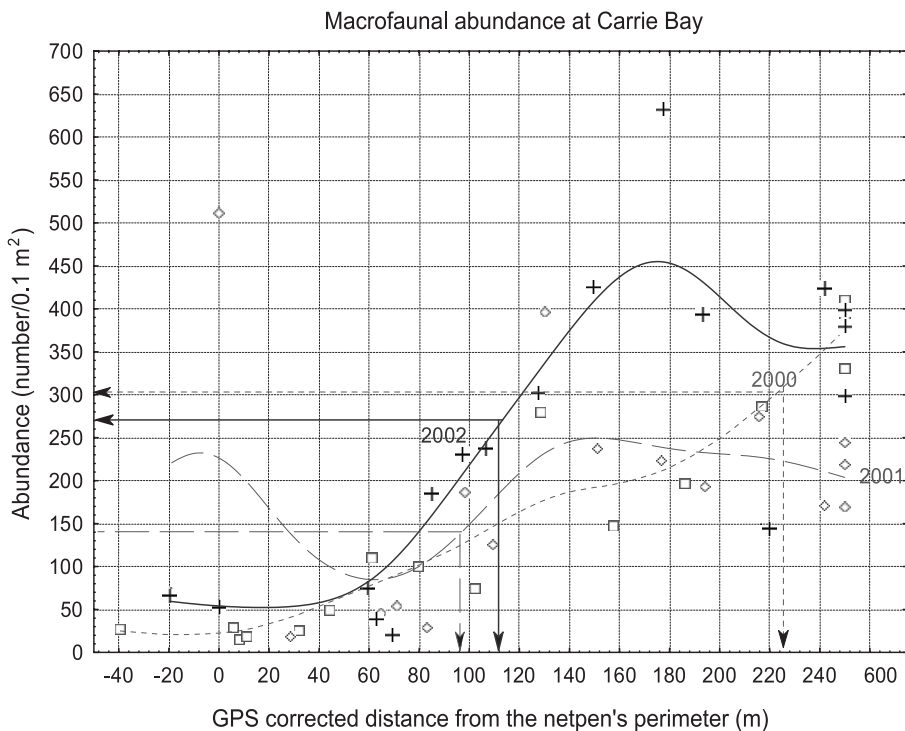


Fig. 9. Abundance of macrofauna observed in 0.1 m^2 van Veen grab samples at the fallow Carrie Bay salmon farm between 2000 and 2002. Horizontal lines represent the lower 90th percentile abundance at the reference station on each sample day. Vertical lines indicate the distance at which observed abundances were equal to the lower 90th percentile. Note the discontinuity in the ordinate. Reference samples were collected at 600 m from the netpen's perimeter on a bearing orthogonal to the prevailing currents.

Carrie Bay, and it is possible if not likely, that speeds >9.5 cm/s were achieved for short periods during each lunar cycle, particularly during spring tides. Fig. 3 (TVS) and 6 (S^-) suggest some lateral dispersion of waste—particularly in 2002. Higher sediment zinc concentrations were recorded at distances between 60 and 150 m to the south from Carrie Bay in 2003 supporting a hypothesis that waste may have been redistributed from beneath the farm to distances of ca. 150 m south during the 5 years in fallow. There is no evidence of waste transport beyond 150 m in any of these graphs and if there was a lateral redistribution of waste at Carrie Bay, it occurred slowly and appeared to have been very limited in extent.

4.4. Macrofaunal history at Carrie Bay treatment stations

The sediment's grain size distribution changed significantly with approach to the southern shore of Carrie Bay. Increasing proportions of shell hash, gravel, cobble and rock were encountered beyond 185 m. The macrofauna at these more distant stations appeared unaffected by farm waste and macrofauna from the 205 and 225 m stations were excluded from the multidimensional analyses because the substrate was very different from either

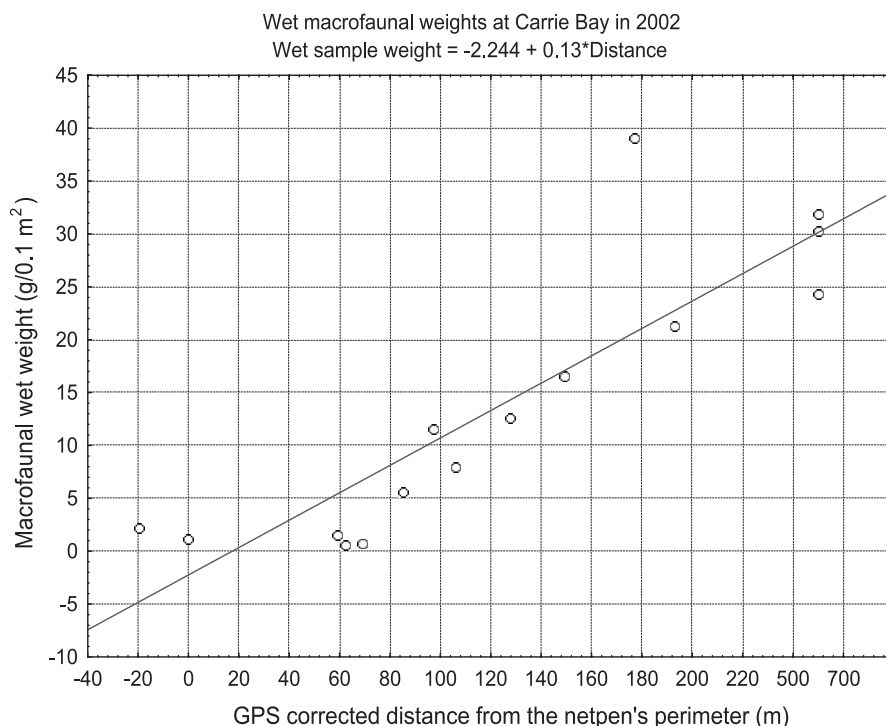


Fig. 10. Macrofaunal wet tissue weights from 0.1 m^2 samples collected at Carrie Bay in 2002. Note the discontinuity in the ordinate. Reference samples were collected at 600 m from the netpen's perimeter on a bearing orthogonal to the prevailing currents.

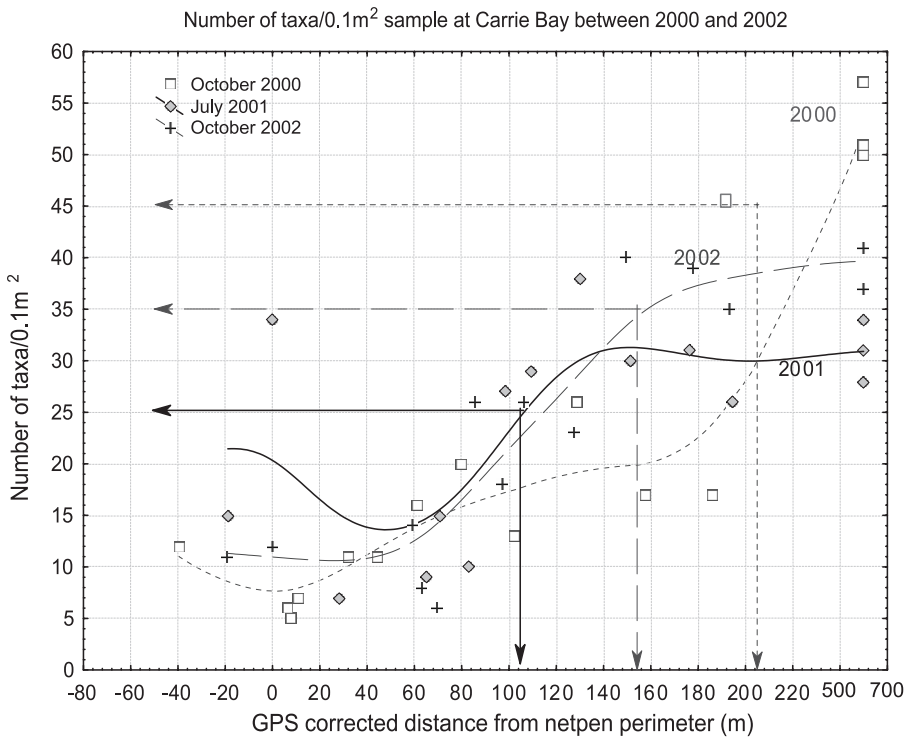


Fig. 11. Number of taxa observed in Carrie Bay farm sediments within 185 m of the netpen's perimeter and at the reference stations. Distance weighted least square fits to the data are provided. Note the discontinuity in the ordinate. Reference samples were collected at 600 m from the netpen's perimeter on a bearing orthogonal to the prevailing currents.

the reference stations or the closer farm stations. Summary physicochemical and biological statistics for all Carrie Bay treatment stations samples collected between 1997 and 2002 are provided in Table 5.

Table 6
Carrie Bay reference area macrobenthic taxa representing $\geq 1.0\%$ of the total abundance

Mollusks	Proportion	Annelids	Proportion
<i>Axinopsida serricata</i>	0.389	Spionidae (3 taxa)	0.048
<i>Nucula tenuis</i>	0.135	<i>Scalibregma inflatum</i>	0.030
<i>Cooperella subdiaphana</i>	0.094	<i>Lumbrineris</i> sp. (4 taxa)	0.028
<i>Acila castrensis</i>	0.030	<i>Chaetozone setosa</i>	0.019
<i>Lucinoma annulata</i>	0.012	<i>Ophelina acuminata</i>	0.019
<i>Alia gaussapata</i>	0.010	<i>Nephtys cornuta</i>	0.015
<i>Macoma secta</i>	0.010	<i>Brada villosa</i>	0.014
<i>Yoldia scissurata</i>	0.010	<i>Heteromastus</i> sp.	0.013
Total mollusks (eight taxa)	0.690	<i>Peisidice aspera</i> or similar	0.010
No. crustaceans represented $\geq 1.0\%$ of total abundance		Total annelids (nine taxa)	0.196

4.4.1. Abundance of macrofauna

A total of 142 taxa were identified in the 51 biological samples collected at Carrie Bay and its reference stations. The abundances of macrofauna retained on 1.0 mm sieves are summarized in Fig. 9. The horizontal lines represent the lower 90th percentile abundance found in reference sediments during each survey and the vertical lines indicate the distance within which abundance was depressed in comparison with this measure of reference conditions. This distance was 225 m in 2000. The abundance of macrofauna increased to reference values at stations located greater than 95 to 115 m from the netpen's perimeter in 2001 and 2002, but remained depressed at closer stations. Macrofaunal abundance was

Table 7
Summary of macrofauna representing $\geq 1.0\%$ of the total Carrie Bay abundance in 2000, 2001 and 2002

Taxon	Trophic level	October 2000 farm	July 2001 farm	October 2002 farm	Reference
<i>Mollusks</i>					
<i>Axinopsida serricata</i>	2	236	492	713	1108
<i>Cooperilla subdiaphana</i>	3	133	264	571	267
<i>Acila castrensis</i>	3	102	32	68	85
<i>Dentalium</i> sp.	3	(3)	49	23	(13)
<i>Nucula</i> sp.	3	42	76	257	410
<i>Yoldia scissurata</i>	3	(4)	68	25	28
<i>Macoma secta</i>	2	(5)	47	(10)	28
<i>Odostomia</i> sp.	3	(0)	(16)	24	(13)
<i>Lucinoma annulata</i>	3	(0)	(0)	(8)	33
<i>Alia gaussipauta</i>	3	(3)	(6)	(8)	28
<i>Polychaetes</i>					
<i>Nephtys cornuta</i>	2	181	170	137	42
<i>Scalibregma inflatum</i>	3	(6)	92	30	87
<i>Ophelina acuminata</i>	4	(2)	28	(3)	53
<i>Lumbrineris</i> sp.	2	(6)	58	25	70
<i>Cossura</i> sp.	3	(1)	24	(2)	(1)
<i>Brada villosa</i>	?	(0)	20	(9)	40
<i>Spionidae</i>	2	13	65	(19)	22
<i>Heteromastus</i> sp.	3	(1)	18	24	37
<i>Prionospio</i> sp.	2	(1)	(5)	(19)	92
<i>Chaetozone setosa</i>	2	(4)	(11)	(10)	53
<i>Peisidice</i> sp.	2	(0)	(8)	(7)	28
<i>Crustaceans</i>					
<i>Pinotheridae</i>	NA	16	27	53	30
Abundance of common taxa		759	1549	2045	2568
Total farm station abundance		892	1734	2209	2848
Proportion common taxa		0.85	0.89	0.93	0.90
Number of common taxa		6	16	12	19
present at $\geq 1\%$					
Number of taxa		62	65	64	61

Taxa found in relatively high abundance in July 2001 on the netpen's perimeter and at a distance of 180 m downcurrent in October 2002 are in bold. All values are abundances in number/m². Counts in parentheses indicate that the taxon was present but represented $<1\%$ of the total abundance in that collection.

increased above reference conditions at stations located ≥ 149 m from the netpen's perimeter in 2002.

4.4.2. Wet weight of macrofauna

The wet weight of macrofauna is provided in Fig. 10. Few organisms of small size were recovered from the area under the netpens and to a distance of 70 m in 2002. Beyond that distance, there was a linear response between sample weight and distance, with one exceptionally high weight (39.1 g/0.1 m²) at 177 m distance where 6320 macrofauna/m² and 39 taxa were observed in 2001. The high tissue weight at that station was associated with a multitude of small animals and not with a few large ones.

4.4.3. Macrofaunal community diversity at Carrie Bay

Fig. 11 summarizes the number of taxa observed in 0.1 m² samples at Carrie Bay between 2000 and 2002. The horizontal lines represent the lower 90th percentile number of taxa observed at the reference stations in each survey year. In 2000, the number of taxa was significantly reduced to less than reference values at all stations located at distances ≤ 185 m, but not at 205 m (45 taxa) or at 225 m (71 taxa). In 2001, when fewer taxa were observed at the reference stations (lower 90th percentile=25 taxa), significant reductions

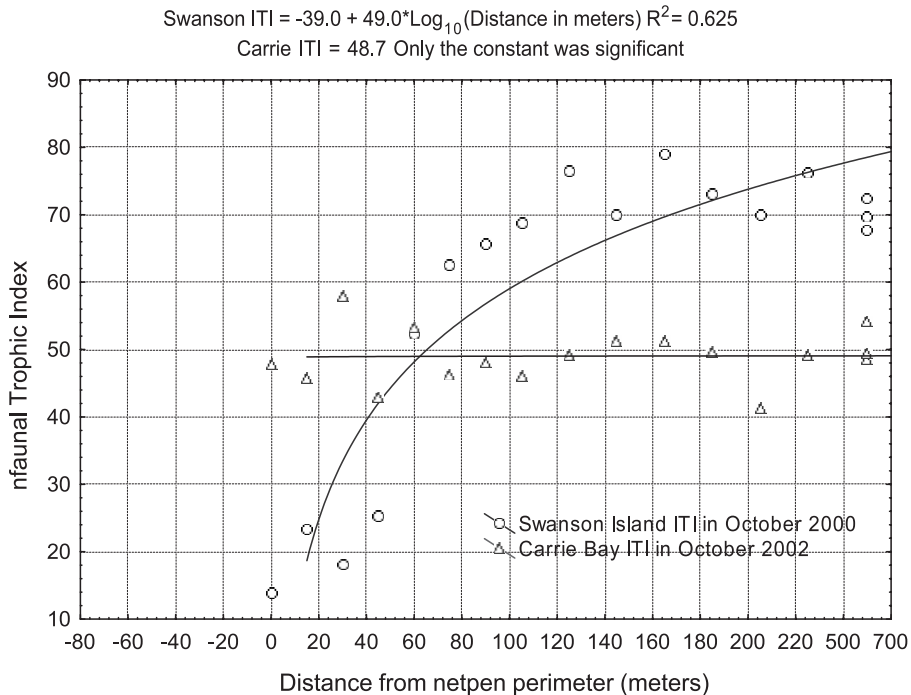


Fig. 12. Infaunal Trophic Index at Carrie Bay and Swanson Island in the fall of the year as a function of distance from the farms. Note the discontinuity in the ordinate. Reference samples were collected at 600 m from the netpen's perimeter on a bearing orthogonal to the prevailing currents.

were observed only within 100 m of the netpen's perimeter. More taxa were observed on the netpen's perimeter and at distances ≥ 105 m in 2001 than were represented by the lower 90th percentile of the number of taxa found in the reference areas.

The 17 common taxa representing $\geq 1.0\%$ of the total abundance on each sample day collected at distances ≤ 185 m from Carrie Bay's netpen perimeter are listed in Table 6. The remaining 119 taxa were far less common, as shown in the *Proportion common taxa* entries. These common taxa, representing only 19% of the number of taxa identified, accounted for between 85% and 93% of the animals identified in the surveys.

Trophic level assignments provided in Table 7 are from Brooks (in preparation) and Word (1990). The majority (19 of the 20 taxa for which trophic assignments were available) of these common taxa were *Surface Detritus Feeders* (Trophic Group 2) or *Surface Deposit Feeders* (Trophic Group 3). *Suspension Feeders* (Group I) were absent and *Subsurface Deposit Feeders* (Group IV) were represented only by the polychaete *Ophelina acuminata*. Brooks (in preparation) and Cromey et al. (2002) have demonstrated the utility of the ITI in assessing the biological response to organic inputs from numerous netpen operations. Fig. 12 provides values of the ITI at Carrie Bay in 2002 and at the Swanson Island salmon farm (Brooks, in preparation) as a function of distance from the netpens' perimeters. ITI values similar to those reported in 2002 were observed in all years evaluated at Carrie Bay. The lack of TVS tolerant opportunistic taxa and of filter feeders

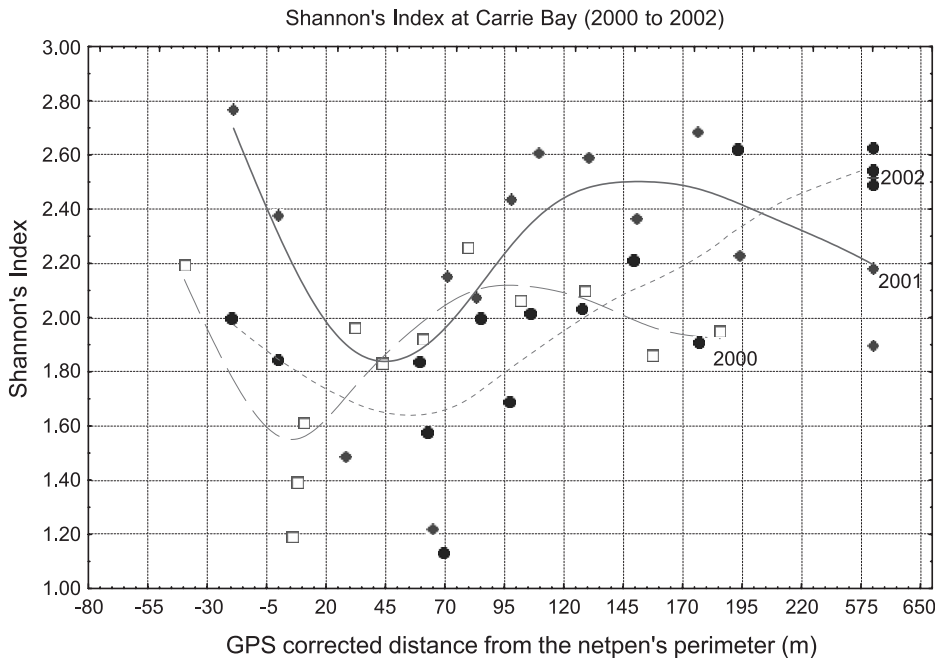


Fig. 13. Values of Shannon's Index for the macrofaunal community found in sediments near the fallow Carrie Bay salmon farm in 2000, 2001 and 2002 as a function of distance from the fallow farm's netpen perimeter. Note the discontinuity in the ordinate. Reference samples were collected 600 m from the netpen's perimeter on a bearing orthogonal to the prevailing currents.

(Group I) in the Carrie Bay environment resulted in nonsignificant trends (the probability that the coefficient on distance equalled zero was 0.50 in 2000, 0.65 in 2001 and 0.83 in 2002) as a function of distance from the netpens and the ITI was not a useful metric for assessing biological effects at this site. In all other instances studied (Brooks, in preparation), the trend in ITI as a function of the biological endpoints studied, which all covaried with distance, was significant. For instance, at Swanson Island (Fig. 13), ITI varied significantly with distance from the farm ($p < 0.000$ and $R^2 = 0.625$). Therefore, the constant ITI values observed in Carrie Bay should not necessarily be interpreted as a limitation on the usefulness of the ITI, but rather as a statement of the uniqueness of Carrie Bay.

Significant increases in abundance above reference conditions were observed under the netpen's perimeter in 2001 and at 180 m distance in 2002. The uncharacteristically abundant samples were dominated by the bolded taxa in Table 7. A total of 6320 organisms/m² were observed at 180 m in 2002. Mollusks represented 5670 of those; annelids 480 and crustaceans 130.

Table 8

Factor loadings resulting from a principal factors analysis of 19 biological and 2 physicochemical variables assessed in 2000, 2001 and 2002 in sediments near the fallowed Carrie Bay salmon farm

Variable	Factor loadings (Varimax raw) (Carrie Bay 2000–2002) extraction: principal factors (MINRES) (marked loadings are >0.600000)	
	Factor 1	Factor 2
Date	0.131	0.396
Distance (GPS DISTANCE)	0.515	0.610
Redox potential in mV (Redox)	0.599	0.606
Sample weight in g (TOTWETWT)	0.300	0.831
Sulfides in micromoles (Sulf)	−0.683	−0.505
<i>Chaetozone setosa</i> (Chtse)	0.558	0.491
<i>Heteromastus</i> sp. (Hetmas)	0.585	0.458
<i>Lumbrineris</i> sp. (Lumb)	0.680	0.233
<i>Nephtys cornuta</i> (Nepcor)	−0.204	−0.189
<i>Ophelina acuminata</i> (Ophel)	0.709	0.271
<i>Spionidae</i> sp. (Spion)	0.728	0.359
<i>Scalibregma inflatum</i> (Scalib)	0.815	0.380
<i>Acila castrensis</i> (Acila)	0.105	0.581
<i>Axinopsida serricata</i> (Axino)	0.614	0.678
<i>Cooperilla subdiaphana</i> (Coop)	0.410	0.613
<i>Lucinoma annulata</i> (Lucan)	0.172	0.718
<i>Macoma secta</i> (Macsec)	0.731	−0.063
<i>Nucula</i> sp. (Nucula)	0.220	0.798
<i>Yoldia scisurata</i> (Yoldia)	0.510	0.182
<i>Alia gaussipauta</i> (Alia)	0.319	0.502
Number taxa/0.1 m ² (TTAXA)	0.742	0.621
Macrofaunal abundance/0.1 m ² (TABUND)	0.622	0.717
Number of mollusks/0.1 m ² (mollusks)	0.550	0.744
Number of annelids/0.1 m ² (annelids)	0.786	0.411
Explained variation	7.440	6.963
Proportion of the total variation	0.310	0.290

Shannon's Index is summarized by year and station in Fig. 13. Values less than 1.8 were determined for communities located only at distances less than 10 m in 2000. Consistent with the hypothesized nearfield advection of farm waste during the fallow period, this zone of low index values moved south about 50 m in 2001 and 2002. These biological surveys started 2 years into the fallow period and higher values of Shannon's Index were observed at the most northerly "upcurrent stations" that were previously located under the netpen's footprint.

4.4.4. Factor analysis

Selected physicochemical and biological variables were transformed and assessed using the Iterated Communality routine in Statistica™. First, multiple R^2 estimates are used for the communalities. After the initial extraction of factors, the method adjusts the loadings over several iterations and evaluates the goodness-of-fit of the resulting solution in terms of the residual sums of squares. The coordinates were then rotated to maximize the explained variation using the Varimax raw routine. As many as five factors were analyzed. However, the two factor analysis presented herein contained nearly all of the pertinent

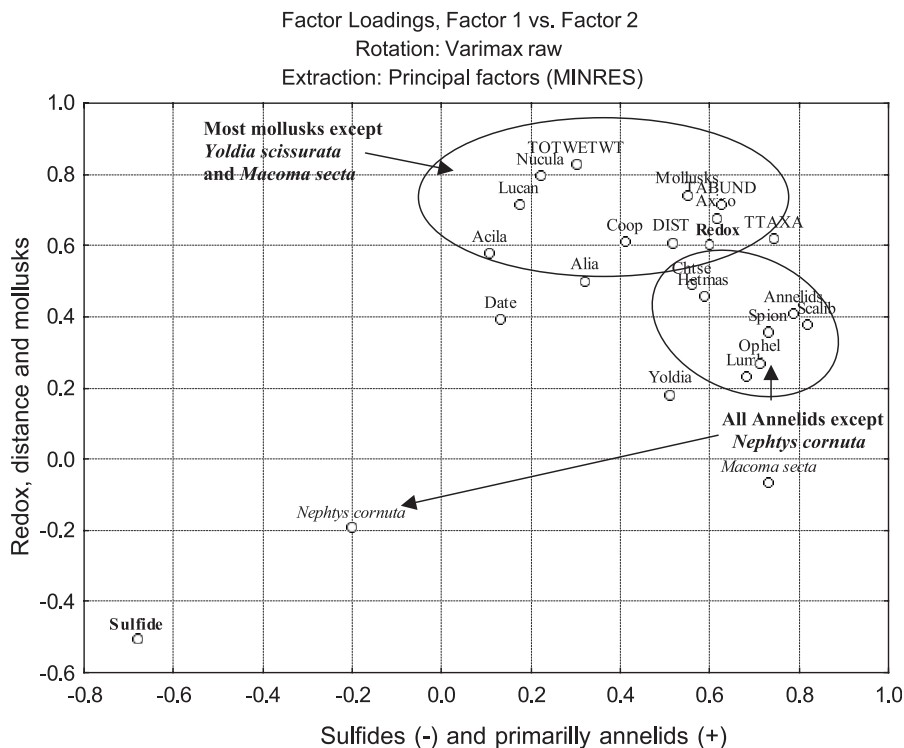


Fig. 14. Graphical summary of the Factor Loadings resulting from a Principal Factors analysis of 19 biological and 2 physicochemical variables assessed in 2000, 2001 and 2002 in sediments near the fallowed Carrie Bay salmon farm.

information. Factor loadings resulting from this analysis are provided in Table 8 where loadings >0.60 are bolded. Free $S^{=}$ were negatively loaded on Factor 1, which explained 31% of the total variation. Significant variables on Factor 1 also included the annelids (*Lumbrineris* sp., *Ophelina acuminata*, *Scalibregma inflatum*, and Spionidae) and two mollusks (*Macoma secta* and *Axinopsida serricata*). These taxa were strongly associated with total abundance, the number of taxa in each sample and the number of annelids in each sample. Sulfides were also negatively loaded on Factor 2 (-0.505), which included positive loadings for redox potential and distance—indicating that redox increased and $S^{=}$ decreased with distance. Significant biological variables were all positively loaded on Factor 2 demonstrating the dependence of sample weight, mollusks, total abundance and diversity on higher redox potentials and reduced $S^{=}$ concentrations. Neither individual annelid taxa, nor the number of annelids in these samples were significant components of any factor in this analysis. Factor 2 explained 29% of the data's variability and together, Factors 1 and 2 explained 60% of the variation. A third factor, significantly loaded only for the proportions of sand and fines was considered for inclusion because it had an eigenvalue of 2.66. However, factor loadings for all other variables were <0.38 and poor associations were found between sediment grain size and other physicochemical or biological endpoints.

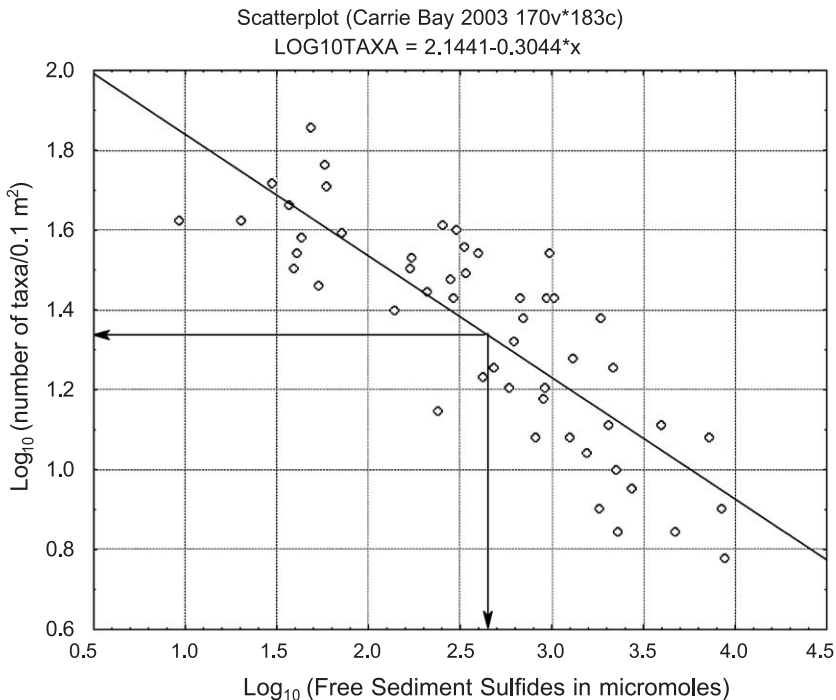


Fig. 15. The number of macrofaunal taxa observed in 0.1 m² van Veen grab samples sieved on 1.0 mm screens at Carrie Bay in 2000, 2001 and 2002. A 50% reduction in the maximum number of taxa observed in these surveys occurred at a sulfide concentration of 447 $\mu\text{M } S^{=}$.

The results are provided graphically in Fig. 14. A key to the variable codes is provided in Table 8. Free $S^{=}$ are distant from all biological endpoints excepting the annelid *Nephtys cornuta*. All other variables group together. The total abundance (ABUND) is closely associated with redox potential (Redox), increasing distance from the farm, the cumulative abundance of all mollusks, and with the most common bivalves (*A. serricata* (Axinop) and *Cooperilla subdiaphana* (Coop)). Total sample wet weight (TOTWETWT) was associated with the largest mollusks in these samples (*Nucula* sp. and *Lucinoma annulata*). Excepting *N. cornuta*, all of the annelids are found together in the highlighted cluster. The intermediate location of the total number of taxa (TTAXA) suggests that it was dependent on both mollusks and annelids.

Concentrations of free sediment $S^{=}$ and redox potential appear to be the driving factors determining the overall relationship between sediment physicochemical and biological endpoints. As previously discussed, the small difference in sediment grain size distribution across the restricted area of these analyses was not strongly associated with any biological variable—except that *N. cornuta* appeared to prefer finer grained sediments. The relationship between $S^{=}$ and the number of taxa observed in these samples is described in Fig. 15. Both variables were transformed ($\log_{10}(N+1)$) for this analysis.

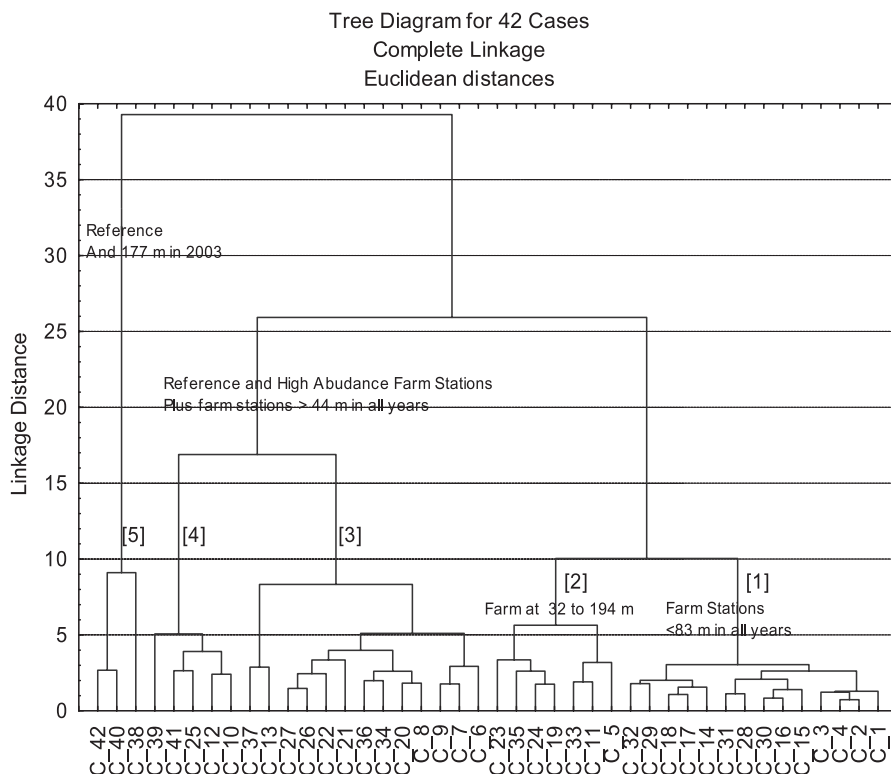


Fig. 16. Tree diagram describing the association of 42 sediment samples representing various distances from the perimeter of the fallow Carrie Bay salmon farm in 2000, 2001 and 2002.

4.4.5. Cluster analysis

The results of cluster analysis are presented in a tree diagram in Fig. 16, which represents data collected in 2000, 2001 and 2002. It does not include physicochemical data collected in 2003 because biological samples were not evaluated in that year. Date and distance were not included in the analysis because these were the independent variables of interest to an analysis of remediation and their inclusion would have resulted in a circular argument. Station membership by date and distance in the primary clusters numbered 1 through 5 are identified in Table 9.

The most highly impacted stations are found in Cluster 1. They include stations under the netpens and to a distance of 10.9 m in 2000. In 2001 and 2002, membership included stations located as far as 83.1 m from the netpens in 2001, suggesting that the most heavily impacted area had moved to the south. In 2002, membership in the most impacted cluster included only those stations located <69.3 m from the netpen's perimeter. Cluster 2 includes intermediate stations located between 32.0 and 194.1 m from the netpen's perimeter. This area of intermediate effects extended further from the farm in 2001 than in other years. However, in 2002, membership in this cluster extended only to 97.3 m. Clusters 3, 4 and 5 all contain reference stations. This suggests that conditions similar to at least some reference samples were found as close as 44.1 m in 2000; 98.1 m in 2001 and 97.3 m in 2002. There is obviously some overlap in membership as a function of distance. For instance, in 2001, intermediate levels of adverse effects were observed to distances as far as 194 m in some samples and other samples, located as close as 109.4 m were more closely associated with reference conditions. This suggests that a mosaic of physicochemical and biological conditions was present in 2001. The 0.0 m station's membership in Cluster 3 will be discussed in a following section of this report.

Table 9

Cluster membership by date and distance (m) associated with the data provided in Fig. 16

Cluster	Date	Distances (m) from the netpen's perimeter of samples in each cluster
1	2000	–39.6, 7.9, 6.1, 10.9
	2001	–18.9, 28.5, 64.7, 70.8, 83.1
	2002	–19.6, 69.3, 62.7, 59.1
2	2000	32.0, 157.4
	2001	98.1, 176.6, 194.1
	2002	85.2, 97.3
3	2000	44.1, 60.8, 79.7, 102.1
	2001	0.0, 109.4, 130.1, 150.9, Reference, Reference
	2002	97.3, 127.6, 149.4
4	2000	128.7, 185.9
	2001	Reference
	2002	Reference
5	2000	177.3
	2001	Reference
	2002	Reference

5. Discussion

Anderson (1992) examined benthic effects at eight salmon farms in British Columbia and found that recovery from moderate to severe impacts could be substantial in times as short as 6 months. Significant improvement in benthic conditions was observed in 1 month at well flushed sites. He observed the proliferation of *Capitella capitata* and *N. cornuta* in highly enriched sediments. While he was unable to conduct long-term studies, he concluded that benthic recovery might take up to 2 years at heavily impacted sites. Mahnken (1993) studied benthic conditions in Clam Bay, Washington State for 2 years and observed complete chemical remediation and the return of all but rare macrobenthic taxa, which remained absent from the vicinity of the farm at the end of the study. More recently, Karakassis et al. (1999) reported improvement in sediment chemistry and all biological endpoints, but noted that neither chemical nor biological remediation was complete at a netpen site in Cephalonia Bay, Greece where sea bream and sea bass had been grown during the preceding 2 years. They noted that the model of Pearson and Rosenberg (1978) did not apply to their site during the study and hypothesized that seasonal variations in benthic chemistry associated with benthic macroalgal production at the shallow site was responsible. Johannessen et al. (1994) studied a Norwegian site with a similar production history and sediment grain size distribution as Carrie Bay. The site, located west of Bergen, Norway produced 2417 mt of Atlantic salmon between May 1988 and May 1990. Sediments on the perimeter of the netpens contained 94% to 97% silt compared with 77% to 78% at the reference station. Reference station TVS was 9.1% to 11.2% and TVS on the perimeter of the netpens varied between 14.0% and 17.3%. Between 30 and 68 taxa were identified at the reference station with a total abundance of between 495 and 1875 macrofauna/m² in the area before culture began. Macrofauna were reduced to five to seven taxa with an abundance of 1935 to 10,370/m² during production. At the end of the study in April 1991, after 1 year in fallow, TVS on the perimeter of the netpens remained at 16.2% and an average of 19 taxa and 21,970 macrofauna/m² were collected in two samples. The high abundance was associated with up to 15,300 *C. capitata* per sample. Thus, while neither chemical nor biological remediation was complete, both factors were improving.

In contrast to the long remediation times reported above, Ritz et al. (1989) observed significant physicochemical and biological remediation under Tasmanian salmon farms in 7 weeks with further improvement 14 weeks post harvest. Lu and Wu (1998) observed biological recovery of the benthos under fish farms at Three Fathom's Cove in Hong Kong within 4 to 5 months and concluded that the fast recovery might have been associated with the subtropical environment, when compared with longer recovery times reviewed in their paper for temperate climes. However, Brooks (1993) found nearly complete chemical recovery and significant biological activity under a Port Townsend, Washington salmon farm 13 weeks following harvest. More recently, Brooks et al. (2003) reported chemical and biological remediation complete at the Arrow Pass salmon farm after 6 months of fallow and chemical remediation was observed and/or predicted to occur rapidly (months) at the Upper Retreat, Midsummer Island and Port Elizabeth salmon farms. Each of these farms produced between 2000 and 2500 mt of Atlantic salmon during their previous production cycles. Thus it appears that rapid benthic recovery can occur in temperate as well as subtropical environments.

The results presented herein indicate that both chemical and biological remediation were progressing at Carrie Bay in 2002 and further reductions in TVS and $S^{=}$ were observed in 2003. However, neither chemical nor biological remediation could be considered complete in May 2003 when the farm had been fallow for 5 years. Based on carbon assimilation modeling using the Findlay et al. (1995) model, Morrissey et al. (2000) predicted long remediation times of 3.1 to 7.5 years under a fish farm at Stewart Island, New Zealand where waste had accumulated to depths of 39.3 cm. The underwater video surveys conducted in 1997 when Carrie Bay was at high biomass and in 1999 after 1 year in fallow revealed waste deposits that qualitatively appeared to vary between <1 and >10 cm deep within 30 m of the netpens. The failure of opportunistic macrofauna to proliferate at Carrie Bay, and other unknown factors, has resulted in slow remediation at this site. However, based on the report of Morrissey et al. (2000) and the observation of accumulations of farm waste in Carrie Bay, it is possible that these accumulations may have contributed to the long remediation time.

5.1. Extent of these effects

Significant benthic effects extended further from the netpens at Carrie Bay than at any of the other 44 farms the author has monitored in British Columbia and Washington State. Fig. 17 is from Brooks and Mahnken (2003a) describing $S^{=}$ concentrations as a function of distance from a large number of B.C. salmon farm netpens and the percent

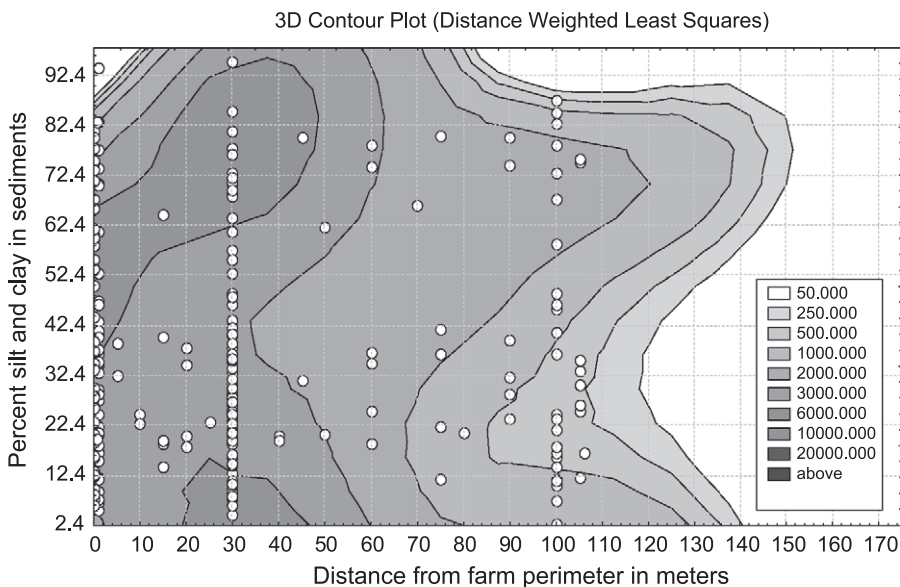


Fig. 17. Contour plot describing $S^{=}$ concentrations (μM) as a function of the percent fines in sediments and distance from the perimeter of 44 salmon farms located in British Columbia.

finer (silt and clay) for values >20% fines. Sulfide concentrations >960 μM , the mean concentration at which Brooks and Mahnken (2003a) reported that the number of taxa declined by one-half, were typically restricted to distances <65 m in sediments containing 25% silt and clay. The extent of these effects increased to 130 m in sediments containing $75 \pm 10\%$ fines, such as those found at Carrie Bay. There are a number of hypotheses that could be invoked in response to these observations. Perhaps dissolved oxygen and sulfate diffusion and infiltration processes are sufficiently reduced in fine-grained sediments to inhibit catabolism of even small amounts of labile organic matter deposited on sediments at distances of 100+ m from salmon farms located in depositional environments. It is also possible that sediments in depositional areas dominated by silts and clays accumulate significant amounts of natural TVS and that they are naturally near their assimilative capacity. In this case, the addition of even small amounts of labile farm waste could result in development of anaerobic conditions. Baseline information was not available for Carrie Bay, but the 9% to 10% TVS observed at the reference stations supports this last hypothesis. It is likely that these and other factors have acted synergistically to reduce TVS catabolic rates in Carrie Bay sediments. The fact remains that this evidence suggests that benthic effects associated with intensive aquaculture may increase with increasing proportions of silts and clays above 55% to 60% fines. Thus, sandy or erosional environments may likely be more suitable sites for intensive aquaculture.

5.2. Chemical remediation

(Fig. 18) describes the history of TVS as a function of time on the perimeter of the Arrow Pass and Carrie Bay salmon farms. Assuming that chemical remediation is tied to catabolism of excess TVS in sediments, the resulting regressions suggest that Arrow Pass would have chemically remediated in 4.8 months of fallow and that TVS in Carrie Bay sediments will be reduced to reference values in 65.1 months. This last prediction must be interpreted with caution because the endpoint lies outside the range of the data. However, the actual remediation time at Arrow Pass is reasonably accurate and the regression suggests that excess TVS will be consumed at Carrie Bay by early 2004.

The TVS data used to construct Fig. 18 was for surficial sediments (upper 2 cm). The data presented in Figs. 4 and 5 suggested a small increase in TVS as a function of depth to 8 cm in a series of sectioned cores collected at Carrie Bay in 2003. The RPD in the immediate vicinity of the farm increased from ca. 1.2 cm in 2000 to 3.1 cm in 2003. The upper 3.1 cm includes those depths where most benthic macrofauna live, suggesting that sediments in 2003 could support new recruits of many species. The increases described in Fig. 4 were small, but they were consistent and statistically significant. These results suggest that microbial catabolism proceeded more slowly at depth in Carrie Bay sediments.

5.2.1. Sulfide

Concentrations of $\text{S}^{=}$ remained moderately high within 65 m of Carrie Bay's netpens during the first 25 months of fallow and then began to decline. This was likely associated with a refractory period during which accumulations of waste created sufficient BOD to maintain anaerobic conditions. Fig. 6 indicates decreasing $\text{S}^{=}$ concentrations at all

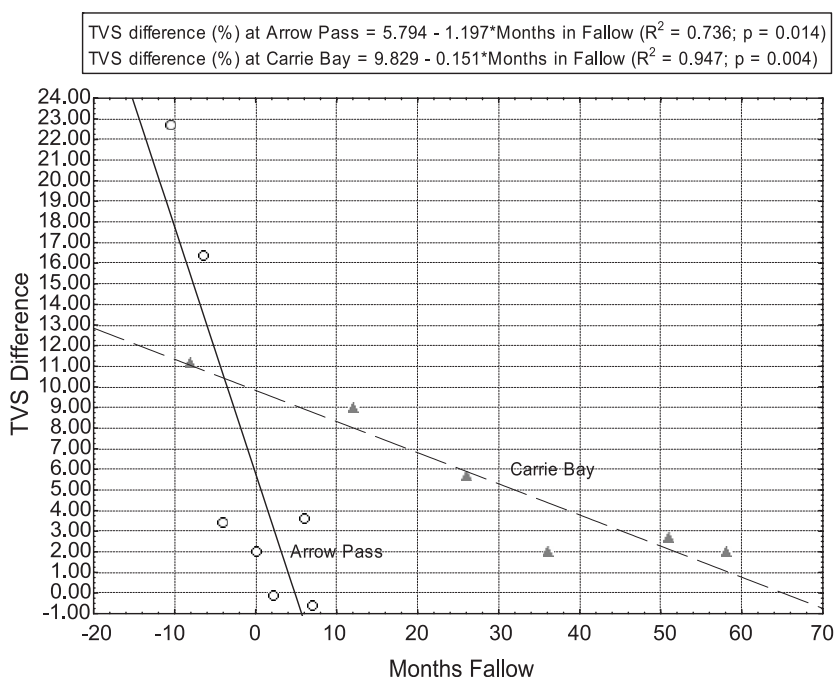


Fig. 18. Organic carbon (TVS) history at Carrie Bay and Arrow Pass in terms of months of fallow. The TVS difference displayed on the abscissa is observed TVS on the netpen's perimeter less the upper 90th percentile TVS observed in all local reference station sediment samples.

distances following 25 months of fallow (August 2000). Fig. 19 describes the $S^{=}$ history in all sediment samples collected at distances <65 m from the netpens perimeter after the 25 month refractory period. Free $S^{=}$ was highly variable in October 2000 resulting in the low coefficient of determination (0.37). However, the coefficient on time (0.0184) was significant at $\alpha=0.05$ indicating that the observed decline was statistically significant. With the caveat that $S^{=}$ remediation is not yet complete, the data suggest that the predicted $S^{=}$ concentration at which half the taxa found at Carrie Bay's reference station can recruit into the area ($447 \mu\text{M}$ described in Fig. 15) will be achieved in surficial sediments at the end of 71 months in fallow (May of 2004). However, the upper 95th CI for $S^{=}$ at Carrie Bay's reference stations ($51.4 \mu\text{M}$ from Table 2) is not predicted to be achieved until the site has been 10 years in fallow (July 2008). Once again, it must be noted that both of these times are in the future and lie outside the range of available data and the predictions are made with caution.

Table 12.1 in Libes (1992) lists oxidation reactions for sedimentary organic matter in marine environments. Possible catabolic pathways, in order of decreasing free energy yields, include aerobic respiration ($\Delta G_w^{\circ} = -3190 \text{ kJ/mol glucose}$) $> \text{MnO}_2$ (-2920 to -3090 kJ/mol) $> \text{NH}_3$ (-2750 to -3030) $> \text{Fe}_2\text{O}_3$ (-1330 to -1410), SO_4^{2-} (-380) and NH_3 via fermentation (-350). The higher free energy yields of aerobic metabolism favors this pathway when compared with sulfate reduction. However, Libes (1992) noted that

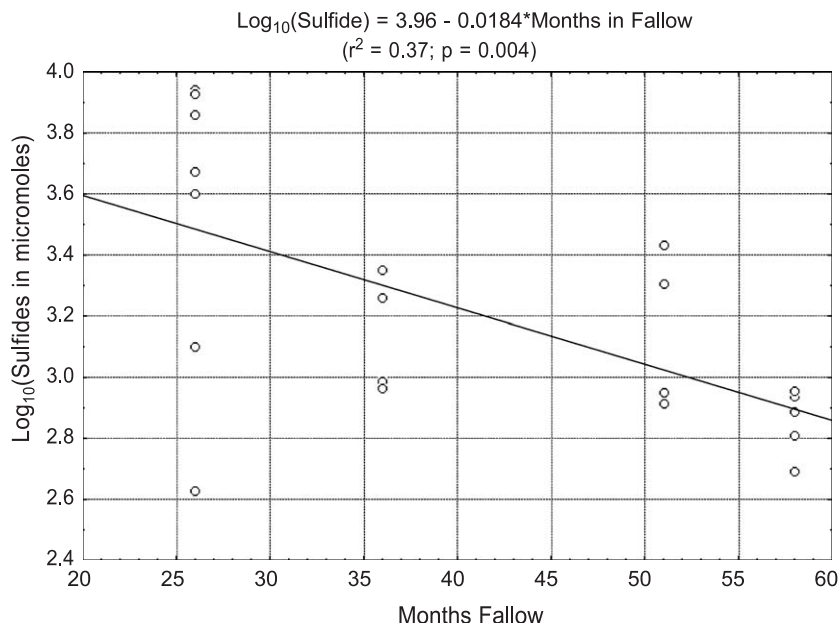


Fig. 19. Sulfide history in surficial sediment (upper 2.0 cm) at distances <65 m from Carrie Bay's netpen perimeter for times representing >25 months in fallow.

only the upper 5 cm of coastal ocean sediments are typically aerobic and Mackin and Swider (1989) observed reducing conditions below a depth 0.2 to 0.25 cm in Long Island Sound sediments. They estimated that oxygen reduction accounted for only 3.3–8.1% of the decomposition of organic matter; nitrate reduction 14.1% to 18.8%; and that sulfate reduction accounted for 73.1% to 82.6%. Other pathways contributed little to the mineralization of organic matter. Assuming that these analyses of more refractory natural TVS can be applied to highly labile salmon farm waste suggests that the sulfur cycle was less efficient at Carrie Bay when compared with Upper Retreat or Arrow Pass. *Beggiatoa*, responsible for oxidizing S^{2-} at the interface between aerobic and anaerobic environments, was not observed within ca. 60 m of Carrie Bay's netpen perimeter in 1997—even though the sediment was black with iron sulfide, emitted gas when disturbed, and appeared to be overlaid by an undetermined depth of waste. The absence of *Beggiatoa* on the sediment's surface suggests that the overlying water was anaerobic and that the bacteria would have been found higher in the water column. Extensive *Beggiatoa* mats were observed at distances >60 m from the farm suggesting that if the overlying water was depleted of oxygen, it was associated with benthic conditions near the netpens and not solely with advection of oxygen deficient bottom water into Retreat Passage. However, this does not preclude the intrusion of bottom water having low oxygen content into this area. In this latter case, a small increase in oxygen consumption by the underlying sediments could deplete water in the benthic boundary layer of already low oxygen concentrations. The absence of *Beggiatoa* in this area that was particularly slow to remediate could have

reduced the efficiency with which $S^{=}$ was oxidized for recycling into sediments sustaining this anaerobic pathway. Other possible factors reducing the rate of anaerobic catabolism in Carrie Bay include reduced diffusion rates caused by the very fine-grained sediments; an excess of divalent cations (Fe^{++} , Mn^{++} , Mg^{++} , etc.) that would compete for $S^{=}$, removing it from the reactive pool. In this environment, it is possible that high iron content would have sequestered significant quantities of sulfide in pyrrhotite (FeS) and eventually in pyrite (FeS_2), neither of which would be detected as $S^{=}$ by the ion specific probes. Sulfur *sinks* such as this could provide an explanation for the reduced $S^{=}$ concentrations observed at Carrie Bay.

5.2.2. Zinc

Di Toro et al. (1992) produced a model indicating that several metals, including zinc and copper, were not biologically available and therefore nontoxic when the molar ratio of simultaneously extracted metal (SEM) to acid volatile sulfides (AVS) was <1.0 . Brooks and Mahnken (2003a,b) reported a maximum free sulfide (FS) to sediment zinc extracted using a strong acid digestion (SADM) of 0.172 for all metal data collected at B.C. salmon farms. The FS:SADM method was found to be a more conservative predictor of biological effects than the SEM:AVS methodology and no adverse effects were predicted in association with sedimented zinc, which is an essential micronutrient in salmon feeds. A maximum FS:SADM ratio of 0.61 was observed at Carrie Bay indicating that the moderately high $S^{=}$ concentrations were sufficient to render the zinc not biologically available and nontoxic.

Cooper and Morse (1998) reported that up to 90% of estuarine pyrite can oxidize within 1 day of exposure to oxic seawater and release the accompanying trace metals. This behavior was reflected in the zinc history at Arrow Pass where Brooks and Mahnken (2003a,b) reported a decline in sediment zinc concentrations from 200 μg Zn/g dry sediment to background (25 $\mu g/g$) during chemical remediation. This same trend was apparent in Carrie Bay sediments (see Figs. 7 and 8) demonstrating that sediment zinc decreased from over 400 μg Zn/g in 1999 to values that with a single exception (237 $\mu g/g$ at 45 m) were less than 120 μg Zn/g in 2002 and 2003. All of these values were less than the Washington State Apparent Effects Threshold (AET) of 270 μg Zn/g or the National Oceanic and Atmospheric Agency AET of 260 μg Zn/g discussed by Brooks and Mahnken (2003b).

5.3. Biological remediation

Mahnken (1993) and Brooks et al. (2003) have shown that following a refractory period of varying length, chemical and biological remediation occurs concurrently. Heilskov and Holmer (2001) found that *Nereis* and *C. capitata* stimulated the mineralization of organic matter by 135% and 87% as measured by O_2 uptake, SO_4^{2-} reduction and the production of CO_2 . While anaerobic processes continued, the presence of both species appeared to primarily facilitate aerobic catabolism. The authors concluded that “Overall, it is clear that the presence of benthic fauna and bioturbation enhances sediment metabolism compared to azoic sediments”. They noted that relatively refractory compounds (like macroalgae and terrigenous plant detritus) are mineralized up to 10 times faster under aerobic than

anaerobic conditions, but that the degradation of more labile compounds (such as fish feces) was unaffected by redox conditions in sediments. Consistent with Kristensen et al. (1995), they suggested that the influence of benthic fauna might be expected to be smaller in fish-farm sediments but asserted the importance of “maintaining macrofaunal populations” to enhance decomposition of organic matter near intensive aquaculture facilities. In contrast, Mackin and Swider (1989) concluded that sulfate reduction was primarily responsible for the decomposition of organic matter in marine sediments and suggested that bioturbation caused by abundant infauna in Long Island Sound sediments had a minor affect on sediment chemistry and the decomposition of organic matter.

The macrofaunal community at Carrie Bay and its reference stations was dominated by mollusks with reduced numbers of annelids and a near absence of crustaceans. The 17 taxa listed in Table 6 represented 88.6% of the total macrobenthos at Carrie Bay’s primary reference location. Brooks (2000, 2001) reported a similar lack of crustaceans and an abundance of mollusks in samples from Arrow Pass and Upper Retreat and this pattern appears consistent in this area of the Broughton Archipelago.

Striplin Environmental Associates (1996) reviewed statistics for macrobenthic organisms in a variety of reference locations in Puget Sound. Table 10 contains a comparison of several macrofaunal endpoints from Puget Sound found in sediments containing between 50% and 80% fines in depths <45.9 m with values observed at Carrie Bay, Arrow Pass and Upper Retreat reference stations. Total abundance was similar at all locations with statistically significantly fewer (*t*-test, $\alpha=0.05$) taxa at Carrie Bay and Upper Retreat in comparison with Puget Sound. The most striking differences are the significantly lower abundance of polychaetes and higher abundance of mollusks at Carrie Bay and Upper Retreat in comparison with similar reference environments in Puget Sound. Table 4 shows that Carrie Bay’s reference macrobenthic community was dominated by mollusks and had fewer taxa and lower total macrobenthic abundance than Puget Sound. Chemical and biological remediation occurred quickly at the Arrow Pass farm and the differences in macrobenthos between this farm and Carrie Bay may be important factors in the longer remediation times. Macrofauna at Upper Retreat, which is located 3 nm from Carrie Bay, was also dominated by mollusks (77% of total abundance) with reduced number of annelids (20%) and a near absence of crustaceans (1.4%). However, chemical remediation has repeatedly been shown to occur within ca. 6 months at Upper Retreat and therefore the

Table 10

Comparison of macrobenthic endpoints observed at Puget Sound reference stations containing between 50% and 80% fines in water depths <45.9 m (Striplin Environmental Associates, 1996) with similar values observed at Carrie Bay’s, Arrow Pass’s and Upper Retreat’s reference stations

Metric	Puget Sound	Carrie Bay	Arrow Pass	Upper Retreat
Abundance/0.1 m ²	343±41 (79)	315±57 (9)	338±114 (6)	497±71 (3)
Number of taxa/0.1 m ²	52±3 (81)	41±6 (9)	59±7 (6)	43±8 (3)
Number of crustaceans/0.1 m ²	51±3 (83)	9±5 (9)	25±19 (6)	7±5 (3)
Number of polychaetes/0.1 m ²	147±15 (82)	87±32 (9)	202±75 (6)	100±12 (3)
Number of mollusks/0.1 m ²	111±27 (78)	217±38 (9)	99±73 (6)	384±51 (3)
Shannon’s Index (<i>H'</i>)	1.23±0.05 (86)	2.38±1.70 (9)	3.23±1.90 (6)	2.39±0.31 (3)
Pielou’s Index (<i>H'</i> / <i>H'</i> max.)	0.74±0.02 (86)	0.70±0.05 (9)	0.81±0.04 (6)	0.64±0.06 (3)

Values are means ±95% confidence intervals with the number of samples (*N*) in parentheses.

presence of a mollusk dominated macrofauna has not been sufficient to predict long chemical remediation times (Brooks et al., 2003). Both chemical and biological remediation were reported complete in 8 months of fallow at the Arrow Pass salmon farm located 5 nm from Carrie Bay, albeit in a different passage. Macrofauna at Arrow Pass were dominated by annelids (60% of total abundance) with fewer numbers of mollusks (29%) and only 7% crustaceans. However, two annelids (*Ophryotrocha vivipara* and *C. capitata*) were observed to proliferate with peak abundances of 6172 and 6756/m² on the perimeter of the Arrow Pass netpens. It should also be noted that mollusks were the last phylum to recruit into Arrow Pass sediments during that 2-year remediation study. This last finding suggests that mollusk dominated macrobenthic communities may take longer to biologically remediate than those dominated by annelids.

No taxa proliferated in Carrie Bay sediments during this study. The maximum abundance of eight taxa found to proliferate in sediments near salmon farms in the Pacific Northwest are compared with their abundance at Carrie Bay in Table 11. *N. cornuta* has been observed in moderate abundance in remediating sediments at several farms and was found in great abundance at Port Townsend in 1993 (Brooks et al., 2003). *C. capitata* is the most commonly found opportunist at many farms, but it was found in only low numbers in Carrie Bay sediments. Opportunistic crustaceans (*Aoroides* sp., *P. oculatus* and *N. pugettensis*) have also been found proliferating at farms located in deep water in Clayoquot Sound, British Columbia. Both Fortune Channel and Bedwell, in Clayoquot Sound, have complex substrates with large amounts of natural marine and terrigenous detritus in addition to moderate amounts of farm waste that had not accumulated to any visible depth. Five and six opportunistic taxa have been observed proliferating at these Clayoquot Sound farms during a single fallow period. Brooks (unpublished) has observed *N. pugettensis* proliferating in subtidal areas of Puget Sound associated with large accumulations of macroalgae and eelgrass detritus suggesting that refractory organic matter may be necessary for the proliferation of these crustaceans.

As seen in Figs. 9, 11–13, biological recovery at Carrie Bay appears to be proceeding from the south toward the netpens and is nearing completion at distances $\geq$ ca. 80 m. Little biological remediation has occurred inside this distance. Sediments within 80 m experienced significant chemical remediation in 2003, but biological samples were not collected in that year. Based on the TVS and S⁼ records, significant biological remediation is expected during the summer of 2003 and in 2004 at all distances. However, biological remediation is apparent in the increasing diversity observed during the fallow period. This can be seen in the number of common taxa representing $\geq 1\%$ of the benthic community at farm stations. In 2000, only six common taxa were found in abundances $\geq 1\%$ of the samples' total. In 2001, there were 16 taxa in this category and in 2002 the number remained twice as high at 12. The number of common taxa observed after 2000 is lower but similar to the 19 common taxa observed at the primary reference stations during these surveys.

Biological recovery has not proceeded in *Stanzas* as described by Mahnken (1993) at Clam Bay in Washington State or by Brooks et al. (2003) for Arrow Pass. Rather, biological remediation has been proceeding through significant recruitment of a single annelid species (*N. cornuta*) and several molluscan taxa (*A. serricata*, *C. subdiaphana* and *Nucula* sp., all of which were common at the reference station and have previously demonstrated an intermediate (MST $\sim$ 2000 μ M S⁼) tolerance for S⁼. The S⁼ tolerance of

Table 11
Abundance (number/m²) of opportunistic taxa observed proliferating in sediments near salmon farms in the Pacific Northwest and their highest abundance at Carrie Bay

Farm	<i>Capitella capitata</i>	Schistomeringos sp.	<i>Ophyrotrocha cf. vivipara</i>	<i>Sigambra tentaculata</i>	Nephtys cornuta	Pseudotanaïs oculatus	Aoroides sp.	<i>Nebalia pugettensis</i>
Port Townsend ^a					7884			
Larsen Island ^b	26,800	30	0	0	0	0	190	0
Swanson Island ^b	3340	20	0	0	130	0	130	40
Bedwell ^b	55,650	17,773	0	7140	170	30	13,310	169,280
Fortune Channel ^b	22,670	10,910	0	1590	0	32,520	6090	12,960
Midsummer Island ^b	2080	20	0	0	470	0	10	50
Upper Retreat ^b	60	0	0	0	250	0	10	0
Arrow Pass ^a	6030	50	1390	0	140	0	140	0
Carrie Bay	220	170	0	0	440	0	20	0

^a Brooks et al. (2003).

^b Brooks (2001).

common taxa found in the vicinity of Carrie Bay are summarized in Table 12. A comparison of the observed $S^{=}$ concentrations described in Fig. 6 with the Minimum Effects Level (MEL) from Brooks (2001) for these taxa suggests that with the exceptions of two annelids (*N. cornuta* and *S. inflatum*) and a single mollusk (*Acila castrensis*), $S^{=}$ concentrations were too high at distances <80 m from the netpens to support substantial recruitment of common taxa until late 2002 or early 2003. Brooks et al. (2003) reported a reduction of 50% in the number of taxa observed at seven British Columbia salmon farms when $S^{=}$ reached 960 μM . Fig. 15 indicated that the number of taxa in Carrie Bay sediments was reduced by half at 447 μM $S^{=}$. The difference in these assessments is tied to the uniqueness of Carrie Bay's biological remediation in which no $S^{=}$ tolerant opportunists proliferated and where the most common taxa had only moderate $S^{=}$ tolerance (Table 12). The relative $S^{=}$ intolerance of Carrie Bay macrofauna may have increased the chemical remediation time and certainly has increased the biological remediation time.

Table 13 describes the abundance of the three most common animals in Carrie Bay as a function of distance and survey date. The abundance of these taxa was depressed to a distance of between 105 and 125 m in 2000. It appears that these taxa signal substantial biological remediation at distances >75 m in 2001 and >60 m in 2002 when their abundance near the farm was depressed but somewhat variable.

5.3.1. Active intervention

Oconnor et al. (1993) described the beneficial effects of harrowing sediments during fallow periods at salmon farms. However, Crema et al. (2000) reported natural recovery of the macrobenthos in the Comacchio lagoon on the northwestern coast of the Adriatic Sea. The 11,500 ha lagoon had been highly enriched during years of eel culture since 1976. The area was fallowed in 1992. Nearly complete biological recovery was reported by the authors based on sampling conducted in 1994. Thus it appears that chemical and biological remediation of this poorly flushed lagoon occurred in 2 years. Based on the

Table 12
Sulfide tolerances of common taxa found in Carrie Bay sediments between 2000 and 2002

Taxon	MEL ^a	HEL ^b	MST ^c	MST (this study)
<i>Axinopsida serricata</i>	725	1300	2300	2145
<i>Cooperilla subdiaphana</i>	245	380	1850	2273
<i>Acila castrensis</i>	2250	2250	2250	8505
<i>Nuculana minuta</i>	300	620	1200	8505
<i>Yoldia scissurata</i>	925	1875	1875	2250
<i>Capitella capitata</i>	4000	7200	7200	7225
<i>Nephtys cornuta</i>	3600	3600	3600	8500
<i>Scalibregma inflatum</i>	540	540	540	1021
<i>Lumbrineris sp.</i>	480	750	3900	969
Spionidae	NA	NA	NA	1021

^a MEL from Brooks (2001) is the highest concentration (μM $S^{=}$) at which a taxon's abundance was consistently found to be elevated above the upper 95th percentile confidence interval found in reference station samples that included the taxon.

^b HEL is the highest concentration of $S^{=}$ where the taxon was found in an abundance $\geq$ the lower 95th percentile of its abundance in all reference station samples where it was found.

^c MST is the maximum $S^{=}$ concentration at which the taxon was found alive.

Table 13

Abundance of the three most common animals (*A. serricata*, *C. subdiaphana* and *Nucula* sp.) as a function of distance from the Carrie Bay netpens and survey year

Year	Distance (m) from netpens												Ref.
	0	15	30	45	60	75	90	105	125	145	165	185	
2000	4	2	7	10	33	20	ND ^a	37	171	93	ND	103	111
2001	158	ND	1	ND	3	6	84	46	170	131	87	111	136
2002	20	ND	4	42	18	123	182	179	224	308	504	202	215

The farm had been fallow for 2 years in 2000 and was not restocked.

^a ND indicates that there was no GPS corrected distance within the spatial limits of this cell.

natural resilience of the system, the authors concluded that *drastic technological measures*, such as harrowing, were not warranted for management of the system. Carrie Bay appears to have been a case where some active intervention, such as harrowing, might have been effective in reducing chemical and biological remediation times—if a reduction in remediation time was deemed necessary.

6. Conclusions

The uniqueness of the remediation characteristics at Carrie Bay challenges our ability to predict recovery rates at only a very small percentage of operating salmon farms in British Columbia. Those challenges also present opportunities and a careful analysis of Carrie Bay, Upper Retreat and Arrow Pass sediments may provide clues helping identify sites where long remediation times might be expected. Unfortunately, resolution of the following questions requires a multidisciplinary approach and cannot be answered with the data collected in support of this study. However, an answer to these questions should be considered important to optimal future siting of intensive aquaculture facilities.

- Sulfide concentrations quickly exceeded 20,000 μM at Upper Retreat and Arrow Pass where chemical remediation occurred very quickly as evidenced by decreases in $\text{S}^{=}$ to <1000 μM within a few months. Why did $\text{S}^{=}$ at Carrie Bay remain at moderate levels ($\leq 9410 \mu\text{M}$) for several years?
- Did elevated sediment concentrations of iron, manganese, magnesium, or other divalent cations provide a sink that removed $\text{S}^{=}$ from the reactive pool and diminishing the rate of anaerobic metabolism?
- Were there other factors at Carrie Bay interdicting the cycling of sulfide to sulfate, which drives anaerobic processes in marine sediments?
- Why have carbon opportunists proliferate at some sites and not at others? Macrofaunal bioturbation may be important with respect to ventilation of sediments, but its value in reducing chemical remediation times associated with catabolism of highly labile organic aquaculture waste is largely unknown and somewhat controversial. Understanding why these fauna have not proliferated at Carrie Bay, where $\text{S}^{=}$ concentrations remained at tolerable levels for these species, could be important to future siting of intensive aquaculture operations.

Chemical remediation is nearly complete at Carrie Bay and a further reduction in sediment $S^{=}$ to $\leq 447 \mu\text{M}$ is predicted by 2004. Biological remediation is proceeding, but was not complete at distances $\leq$ ca. 80 m in 2002. Why this area was characterized by low $S^{=}$ and a long remediation time is unknown but is an area worthy of additional research to provide guidance for siting future intensive aquaculture operations. Based on the information at hand, it appears that locating salmon farms over areas with depositional bottoms containing $>70\%$ to 80% fines may create adverse benthic effects that extend to at least 200 m from the netpens and that may take 5 to 6 years for chemical remediation and somewhat longer for complete biological remediation. However, it should be emphasized that Carrie Bay represented the worst case these authors are aware of in the Pacific Northwest and that while chemical and biological remediations at this site have been much slower than reported by Brooks et al. (2003) for four other farms located in the same area, remediation is undeniably proceeding. There has been no evidence of significant accumulation of carbon at depth in sediments and possible advection of waste has not resulted in an expansion of the benthic area originally impacted by this farm. As has been previously reported for Arrow Pass, sediment zinc concentrations declined during chemical remediation at Carrie Bay and there is no evidence of toxicity associated with this metal to benthic organisms at any time during the operation or remediation of this site.

The uniqueness of Carrie Bay's chemical and biological response to farm waste suggests that environmental effects predictions, based on modeling of the dispersion of waste from intensive aquaculture, will be greatly improved by modules that predict the fate of sedimented labile farm waste. None of the published models currently have that capability. It is likely that a reasonable understanding of these processes could be synthesized from the existing literature. However, that synthesis is beyond the scope of this current report.

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